



# The impacts of agile and lean practices on project constraints: A tertiary study



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## ABSTRACT

The growing interest in Agile and Lean software development is reflected in the increasing number of secondary studies on the benefits and limitations of Agile and Lean processes and practices. The aim of this tertiary study is to consolidate empirical evidence regarding Agile and Lean practices and their respective impacts on project constraints as defined in the Project Management Body of Knowledge (PMBOK): scope, quality, schedule, budget, resources, communication, and risk. In this tertiary study, 13 secondary studies were included for detailed analysis. Given the heterogeneity of the data, we were unable to perform a rigorous synthesis. Instead, we mapped the identified Agile and Lean practices, and their impacts on the project constraints described in PMBOK. From 13 secondary studies, we identified 13 Agile and Lean practices. Test-Driven Development (TDD) is studied in ten secondary studies, meanwhile other practices are studied in only one or two secondary studies. This tertiary study provides a consolidated view of the impacts of Agile and Lean practices. The result of this tertiary study indicates that TDD has a positive impact on external quality. However, due to insufficient data or contradictory results, we were unable to make inferences on other Agile and Lean practices. Implications for research and practice are further discussed in the paper.

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## 1. Introduction

The software industry is highly dynamic and competitive. Software organizations need to accommodate frequent changes in their environment, e.g., customer needs, regulations, and technology, to maintain their competitive advantage (Byrd and Turner, 2000). At the same time software organizations need to deliver in shorter lead times, better quality, and with lower budgets. Flexible approaches such as Agile and Lean software development have emerged to provide solutions to this situation (Dybå and Dingsøyr, 2008; Leffingwell, 2007).

Agile software development is a group of software development methodologies, e.g., Extreme Programming (XP), Scrum, Crystal, etc., that focus on delivering working software products in small iterations, being adaptive towards requirement changes, and collaborating closely with the customers (Abrahamsson et al., 2002). Lean software development is a set of principles and tools derived from Lean manufacturing that focuses on removing waste, delaying

decisions as late as possible, and building quality into the product as early as possible (Poppendieck and Poppendieck, 2003).

The growth of interest in Agile and Lean software development is reflected by the large number of research papers published between 2001 and 2010 (Dingsøyr et al., 2012). However, the strength of empirical evidence regarding the benefits of Agile software development is still very low (Dybå and Dingsøyr, 2008). The benefits and limitations of implementing Agile and Lean practices remain in question.

To better understand the benefits and limitations of Agile and Lean practices, we performed a tertiary study, which is a review of secondary studies (Kitchenham and Charters, 2007). A secondary study is a review of relevant primary studies to synthesize evidence pertaining to a specific research question. Meanwhile a primary study is an empirical study that investigates a specific research question (Kitchenham and Charters, 2007), e.g., case study, experiment, etc. A tertiary study can be performed when there are a sufficient number of secondary studies (ibid.). There are different types of secondary studies:

- A Systematic Literature Review (SLR) is “a form of secondary study that uses a well-defined methodology to identify, analyze and interpret all available evidence related to a specific research

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question in a way that is unbiased and (to a degree) repeatable.” (Kitchenham and Charters, 2007).

- A Literature Review (LR) is “the use of ideas in the literature to justify the particular approach to the topic, the selection of methods, and demonstration that this research contributes something new.” (Hart, 1998).
- A Systematic Mapping (SM) is a form of secondary study that “provides an overview of a research area, and identify the quantity and type of research and results available within it.” (Petersen et al., 2008)

The number of secondary studies in Agile and Lean methodologies has increased over the years. Many of the secondary studies report on the benefits and limitations of Agile and Lean methods and practices. These benefits and limitations are often measured with respect to quality, budget, schedule, which are also known as project constraints (Project Management Institute, 2004). With the increasing number of secondary studies in Agile and Lean methodologies, it becomes important to get a good overview of the existing body of knowledge. Researchers can then build on this body of knowledge or seek new avenues of research. Meanwhile, practitioners get an overview that helps them to better understand the benefits and limitations of Agile and Lean practices.

The purpose of this tertiary study is to consolidate findings on the impacts of implementing Agile and Lean practices as reported in secondary studies. First, we provide an overview of what has been studied in secondary studies on Agile and Lean software development. Second, we examine Agile and Lean practices and their respective impacts and summarize the extent of empirical support as reported in the secondary studies. The outcome of this study is a mapping of Agile and Lean practices and their impacts on project constraints together with the number of empirical studies that support them. The outcome of this study can be used by researchers to identify research gaps and by practitioners to make informed decisions regarding the implementation of Agile and Lean practices.

The remainder of this paper is structured as follows: Section 2 gives an overview over related work. Section 3 presents the review steps. Section 4 provides an overview of the found secondary studies on Agile and Lean methodologies. Section 5 presents the results and a detailed analysis of the impacts of Agile and Lean practices on project constraints. Section 6 presents a discussion of the results and Section 7 summarizes and concludes the review.

## 2. Tertiary studies in software engineering

Evidence Based Software Engineering (EBSE, Kitchenham et al., 2004) has its origin in Evidence Based Medicine (EBM, Straus et al., 2005). The goal of EBM is to integrate research evidence with clinical expertise for patients' well-being. Kitchenham et al. (2004) translated the goals of EBM to suit the contexts of software engineering and formulated the goals of EBSE: (1) a common goal for researchers, (2) a means for industry practitioners to make informed decisions about technology or software development methodology adoption, (3) improve dependability of software systems, (4) improve acceptability of software intensive systems, and (5) an input for certification.

EBSE advocates a more systematic approach to the design and execution of secondary studies, as for example in an SLR (Kitchenham et al., 2004). The aim of an SLR is “to identify, evaluate and summarize the findings of all relevant individual studies, thereby making the available evidence more accessible to decision-makers” (Centre for Reviews and Dissemination, 2009). The result of combining multiple relevant individual studies is expected to provide a more reliable and precise estimate of the effectiveness

of an intervention. In our tertiary study, an intervention is an Agile and Lean practice.

When there are many secondary studies in an area such as in software engineering, it is possible to perform a tertiary study (Kitchenham and Charters, 2007). Kitchenham et al. (2009) performed a tertiary study aimed at assessing impacts of SLRs in software engineering. The study highlights a number of issues pertaining to time spent in performing the review and difficulty to assess the primary studies' quality. Kitchenham et al. (2010) then published a follow up tertiary study and found that the number of SLRs are increasing, and that their quality is improving. However, only a few SLRs perform evaluations of the primary studies and provide guidelines for practitioners.

MacDonell et al. (2010) performed a tertiary study to evaluate the reliability of two independent SLRs with a common research question in the area of effort prediction models. The tertiary study shows that although the two SLRs have different review designs and execution, the results are rather consistent. The tertiary study concludes that systematic reviews are a reasonably robust and reliable research method.

Wohlin et al. (2013) performed a tertiary study to evaluate the reliability of two mapping studies performed by two independent teams within the topic of product families testing. Both mapping studies classified the primary studies using the same scheme proposed by Wieringa et al. (2006). Despite using the same scheme, the mapping studies did not share the same classifications. The tertiary study shows that the results of a secondary study depend on the contexts of the secondary study such as the topic being studied, reviewers conducting the study, search strategies, and available data from the primary studies. However, Wohlin et al. also suggest that secondary studies may reach the same general conclusion even if they include different primary studies.

A tertiary study in Global Software Development (GSD) pertaining to GSD risks and risk mitigation strategies was conducted by Verner et al. (2014). Their study indicates that most SLRs in GSD are mainly literature summaries or mapping studies for researchers. Most SLRs do not provide evidence for the GSD risks that they identified. Also, most SLRs do not provide any advice for practitioners.

There exists one tertiary study regarding Agile methods (Hanssen et al., 2011). However, this tertiary study focuses specifically on the applicability of Agile practices in GSD settings and is based on only two secondary studies. It shows that there is a growing trend of implementing Agile methodologies in GSD. Certain Agile practices, such as continuous integration, pair programming, retrospective, and daily stand up meetings are commonly implemented in GSD contexts. The results indicate that most Agile practices implemented in GSD are successful. However, the indicators of success of implementing Agile were not reported in the tertiary study.

Currently, the benefits and limitations of Agile and Lean practices are reported in independent secondary studies. A consolidated view of findings is required to get a full picture of the benefits and limitations of Agile and Lean practices. Such a consolidated view would allow researchers and practitioners to compare the findings and more easily assess the possible impacts of Agile and Lean practices.

## 3. Review method

The aim of this tertiary study is to provide a consolidated view of findings on the impacts of Agile and Lean practices as reported in secondary studies. The consolidated view will provide researchers and practitioners with more information on whether the impacts of Agile and Lean practices are substantiated with

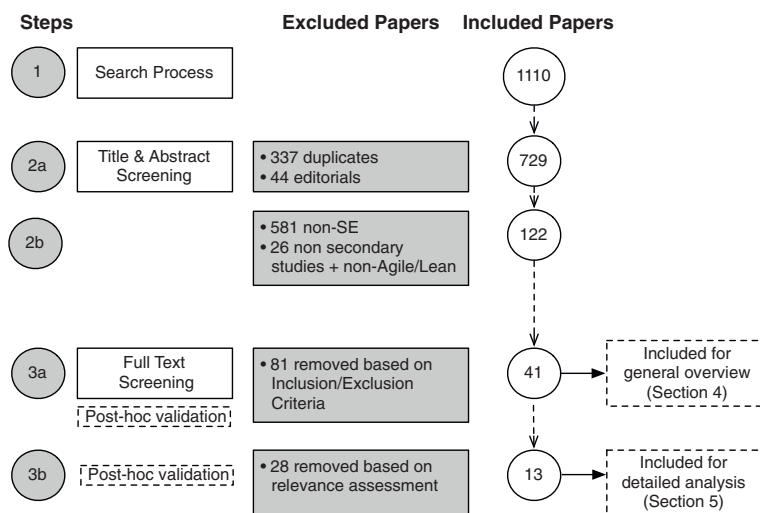


Fig. 1. Steps in the Tertiary Study.

empirical evidence. This will help to better understand the benefits and limitations of Agile and Lean practices.

The following research questions were developed to achieve our aim:

- RQ1:** What are the sets of primary studies that the secondary studies draw upon?
  - RQ1.1:** What is the overlap between the sets of primary studies?
  - RQ1.2:** What are the reasons for overlap and differences?
- **RQ2:** What are the scopes of the secondary studies in evaluating Agile and Lean software development?
  - RQ2.1:** Which Agile and Lean practices were studied?
  - RQ2.2:** What are the impacts of the Agile and Lean practices that were evaluated?
- **RQ3:** What is known about the impacts of Agile and Lean practices?
  - RQ3.1:** What is the level of empirical evidence of the evaluations presented in the secondary studies?
  - RQ3.2:** What is the level of agreement between the secondary studies?
  - RQ3.3:** How does the quality of the secondary studies influence the reported impacts of Agile and Lean practices?

A tertiary study has the same steps as an SLR (Kitchenham and Charters, 2007). In conducting this tertiary review, we adopted the systematic review guidelines by Kitchenham and Charters (2007). Fig. 1 depicts the steps performed in our tertiary study.

### 3.1. Search process: Step 1

The first step (Step 1 in Fig. 1) in our tertiary study was to search for relevant secondary studies. In our search, we formulated search strings that allowed us to retrieve secondary studies in Agile and Lean development from literature databases. The search string was composed of three main substrings: Agile methods, Lean methods, and secondary study. We included Lean methods in the search string because Agile and Lean methods have similar goals of focusing on customers and quickly responding to their needs. Agile and Lean methods also share similar values, principles and practices, and are often considered as similar software development methods (Petersen, 2011). Poppendieck and Poppendieck

**Table 1**  
Keywords in search string.

|             | Substring       | Keywords                                                                                                                                                                                                                                     |
|-------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Substring 1 | Agile           | agile OR scrum OR crystal OR XP OR "extreme programming" OR DSDM OR "dynamic systems development method" OR FDD OR "feature driven development"                                                                                              |
| Substring 2 | Lean            | lean* OR kanban OR kaizen OR "value stream mapping" OR VSM OR "continuous improvement" OR "cross" functional OR "concurrent engineering" OR "integrated product development"                                                                 |
| Substring 3 | Secondary study | "review of studies" OR "structured review" OR "systematic review" OR "literature review" OR "literature analysis" OR "in-depth survey" OR "literature survey" OR "meta-analysis" OR "Past studies" OR "systematic map*" OR "secondary study" |

(2003) also mention that Lean is the extension of Agile's theoretical foundation. Table 1 shows the substrings and the keywords. Surrounding quotes refer to exact matches of a phrase. For single terms, the quotes are redundant and have therefore been omitted.

We performed a pilot search on the databases Inspec and Compendex to check if adding or removing a keyword would lead to more or fewer relevant papers. To formulate the actual search string for Substring 1 and 2, we adapted the keywords that were used in Dybå and Dingsøyr (2008) and Pernstål et al. (2013). We also consulted our search string with Dr. Kai Petersen<sup>1</sup>, who was external to the review team, to validate the keywords included in our search string. For Substring 3, we referred to Kitchenham et al. (2010) to ensure that we included all necessary keywords and synonyms regarding secondary studies. We performed searches on abstract, title, and keywords with the following search string: (Substring 1 OR Substring 2) AND Substring 3.

The search string was then implemented in five electronic databases: (1) Compendex & Scopus, (2) Inspec, (3) IEEE Explore, (4) ACM Digital library, and (5) ISI Web of Knowledge. The search was done in December 2014 to identify peer-reviewed secondary

<sup>1</sup> Dr Kai Petersen is one of the 18 most active researchers publishing in Agile software development, as reported in (Chuang et al., 2014)

**Table 2**  
Database search.

| Database             | Years searched | Total hits  |
|----------------------|----------------|-------------|
| Compendex & Scopus   | 1990 – 2014    | 355         |
| Inspec               | 1990 – 2014    | 414         |
| IEEE Explore         | 1990 – 2014    | 130         |
| ACM                  | 1990 – 2014    | 20          |
| ISI Web of Knowledge | 1990 – 2014    | 191         |
| <b>Total</b>         |                | <b>1110</b> |

studies published since 1990. The search time span was limited to 1990 – 2014 to minimize the number of irrelevant papers from manufacturing, where Agile and Lean concepts were derived from. Table 2 shows the number of papers retrieved from each database.

### 3.2. Study selection: Step 2-3

The following subsections describe the study selection process.

#### 3.2.1. Title and abstract screening: Step 2a – 2b

In study selection, we performed title and abstract screening in two sub-phases. In the first sub-phase of title and abstract screening (Step 2a), we checked for duplicates, as well as non-article papers, e.g., editorials, prefaces, or article summaries. After removing duplicates and non-article papers, we were left with 729 papers. In the second sub-phase of title and abstract screening (Step 2b), we applied an inclusive strategy to avoid premature exclusion of relevant papers. We only excluded papers that were clearly not related to software engineering, no secondary studies, or not in Agile and Lean software development. At this stage only the first author was involved. When in doubt a paper was included. After title and abstract screening, we were left with 122 papers.

#### 3.2.2. Full text screening: Step 3a – 3b

For full text screening, we formulated inclusion/exclusion criteria that we adapted from Kitchenham et al. (2009). The purpose of the inclusion criteria was to ensure that we included all peer-reviewed secondary studies in Agile and Lean software development. Articles were included if they were:

- **IC1.** Systematic reviews, systematic mappings, meta-analyses or literature reviews in Agile or Lean (Type 1) OR secondary studies in Agile and Lean that were part of empirical studies (Type 2).
- **IC2.** Secondary studies (either Type 1 or Type 2) with defined literature review processes. This criterion is adopted from (Kitchenham et al., 2010).
- **IC3.** Secondary studies that were peer reviewed.
- **IC4.** Secondary studies that have a list of included primary studies.

The purpose of the exclusion criteria was to ensure that we only included unique secondary studies with traceable review processes and traceable secondary studies. Articles were excluded if they were:

- **EC1.** Type 2 papers, where the results of the literature studies were not distinguishable from the empirical study results.
- **EC2.** Posters and short papers with less than four pages.
- **EC3.** Redundant reports of the same secondary study.

In the first sub-phase of full text screening (Step 3a), we screened the papers against the inclusion/exclusion criteria. In this stage, we checked the papers for redundancies. In such cases we included the most comprehensive and up-to-date results. For instance, we excluded (Jalali and Wohlin, 2010) from our review and included (Jalali and Wohlin, 2012) instead. We found that the journal version (Jalali and Wohlin, 2012) is an extension of the work

from the mapping study (Jalali and Wohlin, 2010) and also covers more aspects than the mapping study itself. At this stage, the first author reviewed all papers and implemented all inclusion and exclusion criteria. A post-hoc validation was then performed by the second and third authors.

*Post-hoc validation.* A post-hoc validation was performed to ensure that the first author did not exclude papers that should have been included. The second and third authors reviewed two thirds of the 122 papers, where neither authors had overlapping papers. As part of the post-hoc validation, we calculated the Kappa coefficient to evaluate the extent of agreements. The first and second author had a kappa coefficient 0.802, the first and third author had a Kappa coefficient of 0.323. A Kappa coefficient between 0.61 – 0.80 is considered as a *substantial agreement* (Landis and Koch, 1977). Meanwhile, a Kappa coefficient between 0.21 – 0.40 is considered a *fair agreement* (ibid.).

We are aware that a Kappa coefficient could be due to chance (Guggenmoos-Holzmann, 1996). Therefore we examined all papers in detail that were included or excluded by the first author but were included or excluded by the second or third author in a face-to-face meeting. We found that disagreements occurred primarily because the first author was more inclusive than the third author. We did not find papers that were included by the second and third authors but were excluded by the first author. To avoid the exclusion of potentially relevant studies, we continued with the inclusive strategy and kept all papers that were included by the first author. After the first phase of full text screening (Step 3a), 41 papers were left.

In the second phase of full-text screening, we performed a relevance analysis of the secondary studies (Step 3b). In relevance analysis, the first author sought for secondary studies that provide reports, summaries, or statements regarding the impacts of Agile and Lean practices on project constraints. Thirteen out of 41 secondary studies reported an impact of Agile and Lean practices on project constraints.

*Post-hoc validation.* A post-hoc validation was then performed by the second and third authors to ensure that no secondary studies that should have been included were excluded. A discussion was held to discuss disagreements between the first author and the second and third authors. Similar to Step 3a, we found that the first author was more inclusive than the second and third authors. The second and third authors did not include any papers that were excluded by the first author. We, therefore, decided to include all papers that were included by the first author for data extraction.

### 3.3. Data extraction

We extracted the following information from the secondary studies:

1. Meta-data. Title and publication year.
2. Review method. The adopted literature review method and the reference to the methods as mentioned in the secondary study.
3. Aim of study. The aim(s) as stated in the secondary study.
4. Topic of the review. The main focus of the study, whether it is Agile or Lean in general, or specific Agile or Lean practices, or a combination of Agile or Lean with other topics in software engineering.
5. Practices and their associated impacts. The impacts associated with a particular Agile and Lean practice. To ensure uniformity of interpreting Agile and Lean practices, we referred to the list of practices discussed in (Petersen, 2011).
6. Empirical support. The number of primary studies that were cited with respect to impacts and practice association. We also collected the research methodologies that were used in the primary studies



7. List of primary studies. The primary studies that were included in each secondary study.

We extracted items 1–4 from the 41 secondary studies that passed the full text screening (Step 3a). The data extracted from the 41 secondary studies are used to provide an overview of the secondary studies in [Section 4](#). The 13 secondary studies that passed the relevance analysis and provided items 1–7 (Step 3b) are used in the detailed analysis in [Section 5](#). The first author performed data extraction for all 41 papers. A post-hoc validation was performed to check for consistency of the data that were extracted by the first author. The second and third authors performed independent data extractions on ten papers. The data extracted by the first author were then checked and compared with the data extracted by the second and third authors. From the post-hoc validation, we found that data extracted by the first author were also extracted by the second and third author. Therefore, we agreed to use the data that were extracted by the first author.

### 3.4. Quality assessment criteria

Quality assessment is critical because the effectiveness of an intervention might be masked by secondary studies that were not properly conducted ([Centre for Reviews and Dissemination, 2009](#)). For the quality assessment, we adopted the appraisal criteria from the Center for Reviews and Dissemination's (CRD) guide for reviews ([Centre for Reviews and Dissemination, 2009](#)) and adjusted them to the context of this study. The quality assessment criteria are summarized in [Table 3](#).

The quality assessment was performed in parallel with data extraction. We performed quality assessments on the 13 secondary studies that were included for detailed evaluation, because we were interested to examine if the quality of the secondary studies influence the results. We did not exclude any secondary studies based on their quality score.

The first author performed the quality assessment on 13 secondary studies. The second and third author then performed an independent quality assessment of the 13 secondary studies. After the post-hoc validation, we discussed and compared the quality assessments and found only minor deviations in the overall quality scores. From the discussion, we agreed to use the quality assessment that was performed by the first author.

### 3.5. Mapping scheme

Due to the heterogeneity of the data in this tertiary study, we were unable to perform a qualitative meta-study, as described by [Paterson et al. \(2001\)](#). In this study, we decided to aggregate the results using a mapping. In this tertiary study, the secondary studies are clustered based on their quality scores, see [Section 5.3](#). In each quality score cluster, we mapped impacts of an Agile and Lean practice on a project constraint. Each Agile and Lean practice can have a positive, no impact, or a negative impact on a project constraint. We then reported the corresponding number of primary studies, that support a positive, no impact, and a negative impact of an Agile practice on a project constraint. See [Sections 5.4.1–5.4.3](#).

### 3.6. Validity threats

The following are validity threats that are relevant in our tertiary study.

*Missing relevant secondary studies* ([Kitchenham et al., 2010](#)): To mitigate this issue we used a broad search string. Instead of using the individual practices as keywords, we used the Agile and Lean methods in our search. We also omitted “software engineering” or

“software development” in our search string to enable broader hits. We also adopted a more inclusive approach in the study selection. In the title and abstract screening, we included all studies that are on Agile and Lean methods, or secondary studies in software engineering. In the full text screening, discussions were held to examine the disagreements between all three authors. From the discussion, we found that the first author was more inclusive than the other authors. Therefore we agreed to include all papers that were included by the first author, because more papers would not have been included by the second and third authors.

*Missing relevant data from the secondary studies* ([Kitchenham et al., 2010](#)): The first author was the primary reviewer of this tertiary study and thus there was a potential to introduce bias. This issue was mitigated by post-hoc validations performed by the second and third authors for the full text screening, data extraction, and quality assessment. The post-hoc validation process revealed that the second and third author extracted the same data as the first author. The second and third authors did not find more data or different data to be extracted.

*Publication bias* ([Kitchenham and Charters, 2007](#)): Studies with positive results are more likely to be published than those with negative results. This issue did not seem to be a problem as we found secondary studies that reported negative outcomes. There is a risk that the collected secondary studies are skewed towards particular practices. We mitigated this issue by using the Agile and Lean methods, instead of practices, in our search keywords.

*Post-hoc validation*: In other secondary or tertiary studies, usually the included papers are reviewed by at least two reviewers in parallel. In this tertiary study, a post-hoc validation was performed instead. The purpose of the post-hoc validation is to ensure that the main author i.e., the first author, performed a repeatable process of inclusion/exclusion, data extraction, and quality assessment. Such approach is mentioned in ([Brereton et al., 2007](#)) and shown to be helpful to speed up the review process. Performing a post-hoc validation instead of parallel review does not have significant impact on the validity of the study. First, because the results of the study selection performed by the first author were not shared with the second and third author until they completed the post-hoc validation review. Second, all papers were reviewed by at least two independent reviewers; the first author who is a junior researcher, and the second or third authors who are senior researchers. Third, disagreements that arose from the post-hoc validation were resolved through face-to-face discussions. Also, the fact that no papers were reviewed by two senior researchers also did not influence the validity of the study, because as shown in the full text screening step, the first author was more inclusive than the second or third author.

*Quality assessment Tool*: In this tertiary study, we identified a number of simplified versions of an SLR ([Kitchenham and Charters, 2007](#)), they are, semi-SLR ([Mäntylä et al., 2014](#)) and quasi-SLR ([Magdaleno et al., 2012](#)) (definitions of semi-SLR and quasi-SLR are provided in [Section 4.1.1](#)). The authors of these secondary studies provided clear descriptions of the rationale for not performing an SLR and what they have done differently. Despite the clarity of process description provided by the authors, these studies did not obtain high scores in the quality assessment, because they missed a number of steps and therefore obtain low scores for each quality criteria. However, this does not necessarily mean that these secondary studies are bad studies. This is a possible limitation in our quality assessment tool. The quality assessment tool that we adapted from ([Centre for Reviews and Dissemination, 2009](#)) and ([Kitchenham et al., 2010](#)) are tailored for a “complete” SLR. However, not all secondary studies are “complete” SLR. Furthermore, our quality assessment tool is only applicable to assess the report of the secondary studies, and not the actual review process. However, currently, there is no other tool that allows us to assess the

**Table 3**  
Quality assessment score.

| ID  | Criteria <sup>a</sup>                                                                                                           | Score <sup>b</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1  | Was the review question clearly defined in terms of population, interventions, comparators, outcomes and study designs (PICOS)? | Y: PICOS are used and RQs are clearly defined. P: PICOS are not used but RQs are clearly defined. N: RQs are not clearly defined.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| C2  | Was the search strategy adequate and appropriate? <sup>c</sup>                                                                  | Y: The authors have either (searched 4 or more digital libraries <b>and</b> included additional search strategies) or (identified and referenced all journals addressing the topic of interest).<br>P: The authors have searched 3 or 4 digital libraries with no extra search strategies or they searched a defined but restricted set of journals and conference proceedings.<br>N: The authors have searched up to 2 digital libraries or an extremely restricted set of journals.                                                                                                                                          |
| C3  | Are the review's inclusion and exclusion criteria described and appropriate?                                                    | Y: The inclusion/exclusion criteria are explicitly defined in the study.<br>P: The inclusion/exclusion criteria are implicitly defined.<br>N: The inclusion/exclusion criteria are not defined and cannot be readily inferred.                                                                                                                                                                                                                                                                                                                                                                                                 |
| C4  | Were appropriate criteria used to assess the quality of the primary studies? <sup>c</sup>                                       | Y: The authors have explicitly defined quality criteria, quality scoring procedures, and extracted them from each primary study. Information about the quality scores for the primary studies is provided (e.g., in a table or a distribution).<br>P: The research questions involve study quality issues that are addressed by the study. The quality criteria are listed. The quality scoring procedure is described. The scores for individual primary studies are not documented.<br>N: There is no explicit quality assessment of individual primary studies or the quality assessment has been described insufficiently. |
| C5  | Were preventive steps taken to minimize bias and errors?                                                                        | Y: Measures for inter-rater agreement are provided, e.g., Kappa-coefficient(s), or a test-retest (in case of a single reviewer)<br>P: The authors mention preventive actions, e.g., piloting of the review steps.<br>N: Nothing was mentioned pertaining to preventing bias.                                                                                                                                                                                                                                                                                                                                                   |
| C6  | Were adequate details presented for each of the primary studies? <sup>c</sup>                                                   | Y: Information is presented about each paper, so that the data summaries can clearly be traced to relevant papers. E.g., we can trace the research methods, information about study context, or collected measures to individual primary studies.<br>P: Only summary information is presented about individual papers. E.g., papers are grouped into categories, but it is not possible to link individual studies to each category.<br>N: Information/results cannot be traced to individual primary studies.                                                                                                                 |
| C7  | Was the evidence synthesized and aggregated?                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| C7a | Was the evidence actually synthesized and aggregated, or merely summarized?                                                     | Y: The synthesis pools the studies in a meaningful and appropriate way. Differences between studies are addressed.<br>P: A partial synthesis/aggregation of evidence is provided.<br>N: There is no real synthesis. The evidences from the individual studies are basically only repeated/summarized.                                                                                                                                                                                                                                                                                                                          |
| C7b | Were the strengths of evidence of individual studies taken into account in the synthesis?                                       | Y: Yes, to some extent.<br>N: No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| C8  | Do the authors' conclusions accurately reflect the evidence that was reviewed?                                                  | Y: The conclusions are clearly supported by the provided data/results.<br><br>N: The conclusions are not well supported by the provided data/results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

<sup>a</sup> Criteria C1–C8 are adopted from the [Centre for Reviews and Dissemination \(2009\)](#).

<sup>b</sup> Y(es) = 1; P(artially) = 0.5; N(o) = 0.

<sup>c</sup> Details adopted from [Kitchenham et al. \(2010\)](#).

actual review process. To minimize the threat to validity due to the limitation of the quality assessment, we did not use the quality scores to exclude the secondary studies.

**Reliability of the results.** Reliability of the results. [Wohlin et al. \(2013\)](#) discuss reliability issues regarding secondary studies. The outcomes of such studies might depend on the contexts of the reviews, e.g., reviewer experience, review design and protocol, and data available from the secondary/primary studies. However, independent studies may still come to the same general conclusions even if the papers found are different ([Wohlin et al., 2013](#)). In this tertiary study, all authors discussed, formulated and agreed upon the review design and protocol and executed study selection and data extraction accordingly. In our study, the two senior researchers did not review overlapping papers, which might have increased the validity. However, since the same review design and protocol were used by all authors, we believe that reviewing overlapping papers would not have changed the outcome of our study.

## 4. Overview

The results of this tertiary study are divided into two parts. The first part, which we present in this section, is an overview of the secondary studies conducted in Agile and Lean software development. The second part, which we present in [Section 5](#) is the evalu-

ation of Agile and Lean practices and their impacts on project constraints, as well as their empirical support.

### 4.1. Occurrences in agile and lean methods secondary studies

The first part of our analysis is to examine the state-of-art in Agile and Lean secondary studies. The list of included secondary studies is summarized in [Table 4](#).

In total, we found 41 secondary studies: 31 SLRs, four Systematic Mappings (SMs), one literature review, and five secondary studies that were reported as part of an empirical study (Type 2). [Fig. 2](#) shows the trend of published secondary studies.

We did not find any secondary studies that fulfilled our inclusion criteria prior to 2008. The number of secondary studies gradually increased between 2008 and 2012. There was a slight decrease in 2013, but then it increased significantly in 2014. We included a study from 2015 since this paper was published online in 2014, but was assigned to a journal issue in 2015.

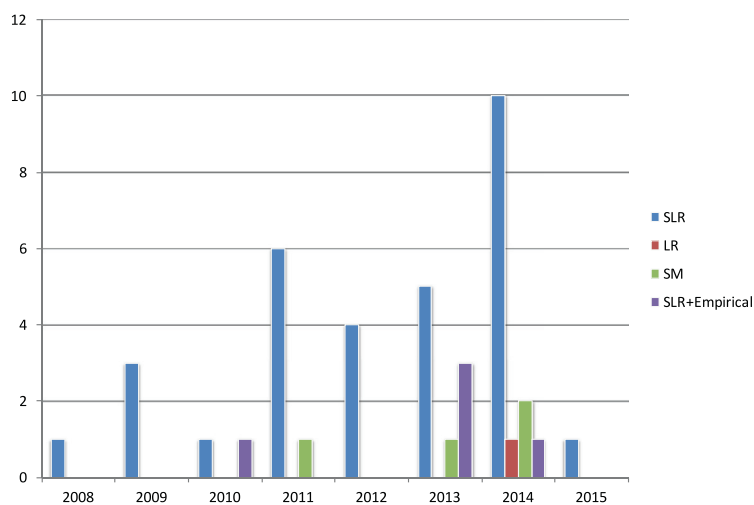
The increasing trend of secondary study in 2008 onward was attributed to the introduction of EBSE by [Kitchenham et al. \(2004\)](#). The low number of secondary studies prior to 2008 could be attributed to the low number of primary studies, as the are was still growing in early 2000. A meaningful secondary study is difficult to be performed with low number of primary studies.

**Table 4**  
List of secondary studies.

| ID  | Citation                       | ID  | Citation                            | ID  | Citation                         |
|-----|--------------------------------|-----|-------------------------------------|-----|----------------------------------|
| S1  | (Ahmad et al., 2013)           | S15 | (Al-Baik and Miller, 2014)          | S29 | (Racheva et al., 2009)           |
| S2  | (Causevic et al., 2011)        | S16 | (Chagas et al., 2014)               | S30 | (Salah et al., 2014)             |
| S3  | (Da Silva et al., 2011)        | S17 | (Silva da Silva et al., 2011)       | S31 | (Salvador et al., 2014)          |
| S4  | (Dybå and Dingsøyr, 2008)      | S18 | (Diaz et al., 2011)                 | S32 | (Senapathi and Srinivasan, 2013) |
| S5  | (Gandomani et al., 2012)       | S19 | (Diebold and Dahlem, 2014)          | S33 | (Silva et al., 2015)             |
| S6  | (Magdaleno et al., 2012)       | S20 | (Hansson et al., 2014)              | S34 | (Sletholt et al., 2011)          |
| S7  | (Munir et al., 2014)           | S21 | (Hebig and Bendraou, 2014)          | S35 | (Stavru, 2014)                   |
| S8  | (Mäntylä et al., 2014)         | S22 | (Hossain et al., 2009)              | S36 | (Usman et al., 2014)             |
| S9  | (Pernstål et al., 2013)        | S23 | (Jalali and Wohlin, 2012)           | S37 | (Yanzer Cabral et al., 2014)     |
| S10 | (Sfetsos and Stamelos, 2010)   | S24 | (Jurca et al., 2014)                | S38 | (Rohunen et al., 2010)           |
| S11 | (Rafique and Mišić, 2013)      | S25 | (Kupiainen et al., 2014)            | S39 | (Stavru et al., 2013a)           |
| S12 | (Shen et al., 2012)            | S26 | (Mitchell and Seaman, 2009)         | S40 | (Stavru et al., 2013b)           |
| S13 | (Zani et al., 2011)            | S27 | (Nguyen-Cong and Tran-Cao, 2013)    | S41 | (Kasoju et al., 2013)            |
| S14 | (Abrantes and Travassos, 2011) | S28 | (Oliveira Albuquerque et al., 2012) |     |                                  |

**Table 5**  
Guidelines used in secondary studies.

| Secondary Study method | Guideline to the method        | Adopted in                                                                                                                                      | Total |
|------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| SLR                    | Kitchenham and Charters (2007) | S1, S2, S4, S5, S7, S10, S11, S12, S13, S15, S16, S17, S18, S20, S21, S22, S23, S25, S26, S27, S28, S31, S32, S33, S34, S36, S37, S38, S40, S41 | 30    |
|                        | Biolchini et al. (2005)        | S17                                                                                                                                             | 1     |
|                        | Pai et al. (2004)              | S14                                                                                                                                             | 1     |
|                        | Unclear                        | S30, S39                                                                                                                                        | 2     |
| Semi-SLR               | Kitchenham and Charters (2007) | S8                                                                                                                                              | 1     |
| Quasi-SLR              | Kitchenham and Charters (2007) | S6                                                                                                                                              | 1     |
| SM                     | Kitchenham and Charters (2007) | S9, S19                                                                                                                                         | 2     |
|                        | Petersen et al. (2008)         | S3                                                                                                                                              | 1     |
|                        | Unclear                        | S24                                                                                                                                             | 1     |
| LR                     | Unclear                        | S35                                                                                                                                             | 1     |



**Fig. 2.** Number of Secondary Studies performed according to Year.

#### 4.1.1. Guidelines used in the secondary studies

We found different methods being employed in the secondary studies. We also found that different guidelines are used in the secondary studies. Table 5 summarizes the methods and guidelines mentioned in the secondary studies. Some papers are mentioned more than once in the table, and mapped into different methods and guidelines. This was done because more than one method and/or guideline are mentioned in their secondary studies.

Most of the secondary studies adopted the SLR guidelines by Kitchenham and Charters (2007). One study (S17) adopted the SLR guideline by Biolchini et al. (2005). The authors of S14 adopted an SLR guidelines from medicine, i.e., Pai et al. (2004). Four secondary studies did not mention which guidelines were used. We also

found modified or simplified versions of Kitchenham and Charters's SLR guidelines being used in the secondary studies, they were, quasi-SLR (S6), and semi-SLR (S8). A quasi-SLR is a term introduced by Travassos et al. (2008) to define a secondary study that follows has the same rigor and formalism of an SLR, but no meta-analysis in principle can be applied. The authors of S8 refer to their secondary study as semi-SLR because it does not follow the same rigour as an SLR, such as, having only one reviewer, no rigorous quality assessment of the study, and no data synthesis.

We also found a number of SMs in Agile and Lean software development. When we examined the year of publications, SMs were gaining popularity as a method for studying Agile and Lean software development literature since 2011.

**Table 6**  
Aims of secondary studies.

| Aim                      | References                                                                                                    |
|--------------------------|---------------------------------------------------------------------------------------------------------------|
| Identify Factors         | S4, S5, S6, S8, S12, S13, S14, S15, S16, S17, S18, S20, S22, S23, S28, S29, S30, S32, S33, S35, S37, S38, S39 |
| Survey the literature    | S1, S25, S27, S31, S34, S36, S40, S41                                                                         |
| Classify primary studies | S3, S9, S19, S21, S24                                                                                         |
| Find Association         | S2, S7, S10, S11, S26                                                                                         |

Most secondary studies follow certain guidelines. This finding suggests the increasing need to gather evidences pertaining to Agile and Lean software development in more systematic ways. We also found two secondary studies with modified or simplified the SLR steps defined by Kitchenham and Charters (2007). The finding indicates the possibility of performing less rigorous secondary study but remain traceable and systematic.

#### 4.1.2. Aims of secondary studies

We identified four types of secondary studies' aims:

- **Classify Primary Studies.** Secondary studies in this category reviewed the primary studies and classified them into different categories. All SM studies fall into this category.
- **Find association.** Secondary studies in this category reviewed the primary studies and examined the association between Agile or Lean practices and their impacts on different aspects, such as quality, schedule, etc.
- **Identify Factors.** Secondary studies in this category reviewed the primary studies to identify various factors, like challenges, benefits, limitations, or certain practices.
- **Survey the literature.** Secondary studies in this category reviewed the primary studies to identify trends and extents of Agile and Lean software development research.

In categorizing the papers, we looked at the stated aim and how the results are presented. Table 6 summarizes the aims of secondary studies. The detailed aim from each secondary study is compiled in Appendix A (Table A.27).

Most secondary studies are aimed to **identify factors** pertaining to Agile and Lean methods, such as commonly used practices, challenges of implementing Agile methods, etc. When we examined the type of secondary study (see Section 4.1.1), most of the secondary studies in this category mentioned, in the title or indicated in the text, to have performed an SLR, except S6, S8, and S35.

We also found a number of secondary studies that are aimed to **survey the literature**. All secondary studies in this category also mention having performed an SLR. Five secondary studies are aimed to **classify primary studies**. All secondary studies in this category, except S21, mention having performed a systematic mapping study. S21 on the other hand mention having performed an SLR. There were very few secondary studies that were aimed **find associations** between Agile and Lean methods or practices and their impacts on cost, quality, productivity, etc. Three secondary studies in find association (S2, S7, S11) are focused on TDD.

#### 4.1.3. Topics in agile secondary studies

We found a variety of topics in Agile and Lean secondary studies. Combinations of Agile and Lean software development with other areas in software engineering, such as Software Product Line Engineering (SPLE) and Global Software Development (GSD), dominate the secondary studies' topics. General discussions on Agile and Lean software development is also a popular topic.

Table 7 summarizes the topics covered by the secondary studies. It shows that Agile and Lean practices are implemented in

**Table 7**  
Topics of secondary studies.

| Topic                                                                                                                                     | References                                                                    | Total |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------|
| Combining Agile/Lean with other topics in SE, e.g., User Centered Design (UCD), SPLE, GSD, Metrics, Model Driven Development (MDD), Cloud | S3, S13, S17, S18, S20, S21, S22, S23, S24, S25, S27, S30, S31, S36, S39, S40 | 16    |
| Agile/Lean General                                                                                                                        | S4, S14, S19, S32, S35, S38                                                   | 6     |
| Application of Agile/Lean in specific domains e.g., Embedded Systems, Scientific Software, Large Scale Software Systems, Automotive       | S9, S12, S28, S34, S41                                                        | 5     |
| Test Driven Development (TDD)                                                                                                             | S2, S7, S11                                                                   | 3     |
| Comparing Agile with other development process models, e.g., Open Source Software (OSS), Plan driven)                                     | S5, S6, S26                                                                   | 3     |
| Lean - Kanban                                                                                                                             | S1, S15                                                                       | 2     |
| Agile and Management (Project Management, Knowledge Management)                                                                           | S16, S37                                                                      | 2     |
| Agile and Value Creation and Quality                                                                                                      | S10, S29                                                                      | 2     |
| Agile and CMMI                                                                                                                            | S33                                                                           | 1     |
| Frequent Releases                                                                                                                         | S8                                                                            | 1     |

a wide range of domains, such as embedded systems, scientific software, and regulated domains. We also found secondary studies that focus on specific Agile and Lean practices, such as TDD and frequent releases. The variation of secondary topics further reinforces the prevalence of the use of Agile or Lean methods and practices.

## 5. Results and analysis

This section presents the detailed analysis of the impacts of Agile and Lean practices on project constraints.

### 5.1. Population of primary studies

We examined the primary studies from the 13 secondary studies. We found 50 overlapping primary studies out of 420 primary studies included in 13 secondary studies. However, the 50 overlapping primary studies are not cited by all 13 secondary studies. Fig. 3 shows the overlap across the secondary studies. The number in the brackets represents the total number of primary studies from each secondary study.

From Fig. 3, we can see four clusters of papers (rounded boxes). **Lean Cluster** contains papers focusing on Lean methods (S1, S9) and frequent deliveries (S8). **Agile/SPLE Cluster** contains papers on the combination of Agile and software product line engineering (S3, S13). The secondary studies in **Miscellaneous Cluster** has a variety of topics, including Agile and Lean software development in general (S4), Agile methods in the context of open source (S5, S6), and implementation of Agile in embedded systems development (S12). Lastly, **TDD Cluster** contains three papers focused on TDD (S2, S7, S11) and a paper focused on Agile methods in general (S10).

Although the secondary studies in **Lean Cluster** shared the same topic, they have little overlap in primary studies. The same pattern is also observed in **Agile/SPLE Cluster**. The small overlap of primary study could be attributed by the difference of focus among the secondary studies in each cluster. For instance in Lean Cluster, S1 is focused on Kanban, and S9 has a broader focus on Lean in general. In Agile/SPLE Cluster, S3 focuses on combining Agile meth-



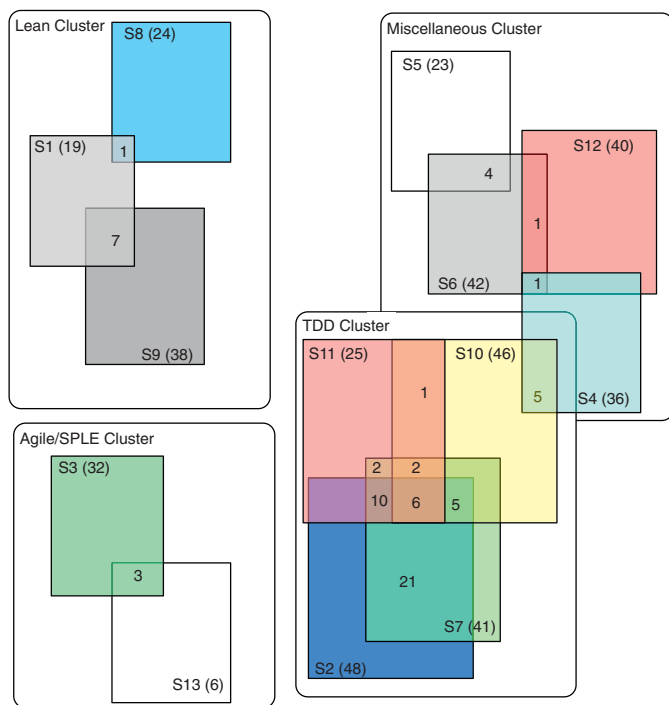


Fig. 3. Overlap of Primary Studies Across Secondary Studies.

Table 8

List of most cited papers from the secondary studies.

| Papers                                                                                                                                                                                                                       | Cited In         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| D. Karlström, P. Runeson, Combining agile methods with stage-gate project management, <i>IEEE Software</i> 22(3) (2005), pp. 43 – 49.                                                                                        | S4, S6, S12      |
| N. Nagappan, E.M. Maximilien, T. Bhat, L. Williams, Realizing quality improvement through test driven development: results and experiences of four industrial teams, <i>Empir. Softw. Eng.</i> 13 (3) (2008), pp. 289 – 302. | S2, S7, S10, S11 |
| L. Williams, E. Maximilien, M. Vouk, Test-driven development as a defect-reduction practice, in: 14th International Symposium on Software Reliability Engineering (ISSRE 2003), pp. 34 – 45                                  | S2, S7, S10, S11 |
| L. Huang, M. Holcombe, Empirical investigation towards the effectiveness of test first programming, <i>Inform. Softw. Technol.</i> 51 (1) (2009), pp. 182 – 194.                                                             | S2, S7, S10, S11 |
| H. Erdogmus, M. Morisio, M. Torchiano, On the effectiveness of the test-first approach to programming, <i>IEEE Trans. Softw. Eng.</i> 31 (3) (2005), pp. 226 – 237.                                                          | S2, S7, S10, S11 |
| M. Muller, O. Hagner, Experiment about test-first programming, <i>IEE Proc. Softw.</i> 149 (5) (2002), pp. 131 – 136.                                                                                                        | S2, S7, S10, S11 |
| A. Gupta and P. Jalote, An Experimental Evaluation of the Effectiveness and Efficiency of the Test Driven Development, <i>Proc. First Int'l Symp. Empirical Software Eng. and Measurement</i> , pp. 285 – 294, 2007.         | S2, S7, S10, S11 |

ondary studies and compared them against the description provided by Petersen (2011). We also used the definitions presented in Williams (2010) as a cross reference to help with identifying the practices.

We collected the impacts of Agile and Lean practices bottom-up from the included secondary studies. Each impact was then mapped to the following project constraints as defined in Project Management Body of Knowledge (PMBOK): scope, quality, schedule, budget, resources, communication, and risk (Project Management Institute, 2004). The mapping of studies to practices and project constraints is summarized in Table 9.

### 5.2.1. Agile and lean practices

From the list of practices described by Petersen (2011), we identified 13 out of 26 practices from the secondary studies. TDD is mentioned in 10 secondary studies. Pair programming and planning game are mentioned in two secondary studies. The remaining practices, on-site customer, refactoring, incremental deliveries, internal/external release, short iteration adaptive planning, stand up meeting, team chooses own task, Value Stream Mapping (VSM), and Kanban pull systems, is mentioned in one secondary study.

Often we found different terms being used in the secondary studies when referring to a practice. Some of the secondary studies use the same terms as in Petersen (2011), which are, *on-site customer*, *refactoring*, *pair programming*, *planning game*, and *VSM*.

The secondary studies that discuss the practice of creating test cases prior to coding is mapped into *TDD & test automation*. We also identified synonyms to TDD, such as test-first programming (S2, S4) and test-first development (S11).

Rapid releases (S8) is mapped into *incremental deliveries*, as it involves releasing new or updated versions of software to the customer frequently (Petersen, 2011). Rapid releases does not fit into the definition of *short iteration* as it entails performing development activities in short periods of time, where a feature was delivered to obtain feedback from the customers. The definitions provided for *short iterations*, both in (Petersen, 2011) and (Williams, 2010), do not entail releasing a version. However *short iterations* fits the definition of a practice that involves delivering in-progress software to obtain continuous feedback mentioned in S1.

The secondary study mapped into *internal/external releases* (S9) discusses a practice of dividing development task into major and

ods and SPLE, and S13 focuses on the use of reference architectures in Agile software development.

In **Miscellaneous cluster**, there are three secondary studies (S6, S10, S12) that have overlaps with S4, because S4 has a general focus on the implementation of Agile and Lean software development. S5 and S6 share four overlapping primary studies because they share the topic of combining Agile methods and open source.

In **TDD cluster**, we identified many overlaps among the secondary studies (S2, S7, S10, S11). S2, S7, S10, and S11 have six overlapping primary studies. Three papers that focus on TDD (S2, S7, S11) have ten overlapping primary studies. The largest overlaps are between S2 and S7, with 21 overlapping primary studies. Given the similarity of focus, we expected more overlap among S2 and S7. However, if we looked closer into S2 and S7, S7 is focused on the impact of TDD on quality and productivity, and S2 is focused on the impact of TDD in general. We did not expect large overlaps between S2 and S7 on one hand and S11 on the other because S11 only included experiments in TDD. The large number of empirical studies conducted in TDD shows that TDD has been well researched. The lack of overlap suggested that researchers in TDD had separate agendas in examining the impact of TDD. Also, differences in secondary study's review design/protocol could also influence the number of overlapping primary studies.

We also found primary studies that are cited in three to four secondary studies. In Miscellaneous Cluster, S4, S6, S12 share one primary study, and in TDD Cluster, S2, S7, S10, S11 share six primary studies. These primary studies could be considered as key papers in Agile and Lean software development. The list of key primary studies is presented in Table 8.

### 5.2. Scopes of agile and lean secondary studies

We used the list of Agile and Lean practices from Petersen (2011) to see which Agile and Lean practices had been examined and which had not. The list from Petersen (2011) was selected as it covers both Agile and Lean practices. To identify the practices, we looked for the description of the practice in the sec-

**Table 9**  
Practices and project constraints mapping.

| Practice                                                                 | Project constraints |                                            |                       |         |             |       |               | Sum |
|--------------------------------------------------------------------------|---------------------|--------------------------------------------|-----------------------|---------|-------------|-------|---------------|-----|
|                                                                          | Scope               | Quality                                    | Schedule              | Budget  | Resources   | Risks | Communication |     |
| Onsite customer                                                          |                     |                                            |                       |         |             |       | S4            | 1   |
| Metaphores and user stories                                              |                     |                                            |                       |         |             |       |               | 0   |
| Refactoring                                                              |                     | S10                                        | S10                   |         |             |       |               | 1   |
| Coding standards                                                         |                     |                                            |                       |         |             |       |               | 0   |
| Team code-ownership                                                      |                     |                                            |                       |         |             |       |               | 0   |
| Low dependency architecture                                              |                     |                                            |                       |         |             |       |               | 0   |
| TDD & test automation                                                    | S12                 | S2, S3, S4, S5, S6, S7, S10, S11, S12, S13 | S2, S7, S10, S11, S12 | S2, S10 | S2, S7, S10 |       |               | 10  |
| Pair programming                                                         |                     | S4, S10                                    | S4, S10               |         | S10         |       | S10           | 2   |
| Continuous integrations                                                  |                     |                                            |                       |         |             |       |               | 0   |
| Inspection                                                               |                     |                                            |                       |         |             |       |               | 0   |
| Configuration Management                                                 |                     |                                            |                       |         |             |       |               | 0   |
| Incremental deliveries                                                   |                     | S8                                         | S8                    |         |             | S8    | S8            | 1   |
| Internal/external release                                                |                     |                                            | S9                    |         |             |       |               | 1   |
| Short iteration                                                          |                     | S4                                         |                       |         |             |       |               | 1   |
| Adaptive planning with highest priority user stories (Adaptive planning) |                     | S1                                         | S1                    |         |             |       | S1            | 1   |
| Time-boxing                                                              |                     |                                            |                       |         |             |       |               | 0   |
| Planning game                                                            |                     |                                            |                       |         | S4, S10     |       | S4            | 2   |
| Co-located development                                                   |                     |                                            |                       |         |             |       |               | 0   |
| Whole team (cross-functional team)                                       |                     |                                            |                       |         |             |       |               | 0   |
| 40h week                                                                 |                     |                                            |                       |         |             |       |               | 0   |
| Stand up meeting                                                         |                     |                                            | S4                    |         |             |       | S4            | 1   |
| Team chooses own task                                                    |                     |                                            | S1                    |         |             |       |               | 1   |
| VSM                                                                      |                     | S9                                         | S9                    |         | S9          |       |               | 1   |
| Inventory management                                                     |                     |                                            |                       |         |             |       |               | 0   |
| Chief Engineer                                                           |                     |                                            |                       |         |             |       |               | 0   |
| Kanban pull systems                                                      |                     | S1                                         | S1                    |         |             |       |               | 1   |
| <b>Sum</b>                                                               | 1                   | 13                                         | 8                     | 2       | 4           | 1     | 4             |     |

minor enhancement. The secondary study mapped into *stand-up meeting* (S4) discusses a Scrum practice of daily meetings that involves different stakeholders.

S1 mentions two Kanban practices, visualization and prioritization of tasks. S1 does not provide a description of visualization. We assumed they are a visual means for tracking and prioritizing tasks, as in a Kanban board. Meanwhile, S1 defines prioritization as a practice that helps in deciding which work item needs to be pulled next to optimize values. Although S1 separated the two practices apart, they both fit the definition for *adaptive planning with highest priority user stories*. To shorten the term, we will refer adaptive planning with highest priority user stories as *adaptive planning* in the remainder of the paper. S1 discusses a Kanban practice where team members are allowed to decide their own task management and prioritization, without upper level management approval. This practice is mapped under *team chooses own task*. S1 also mentions a Kanban practice that limits the work in progress by pulling the next work item according to the team's capacity. We mapped this practice into *Kanban pull systems*.

### 5.2.2. Impacts of agile and lean practices

From Table 9 we can see the impact of Agile and Lean practices on quality is reported in 13 secondary studies. Schedule is reported in eight secondary studies. Resource and communication are reported in four secondary studies. Budget is reported in two secondary studies. Meanwhile scope and risks are reported in one secondary study.

We referred to available guidelines and standards, like PMBOK (Project Management Institute, 2004) and ISO 25010 (ISO/IEC, 2010) to map the impacts of Agile and Lean practices on the relevant project constraints. We mapped fulfilling requirements (S12) into scope because activities in scope management include managing requirements (Project Management Institute, 2004). We

mapped productivity (S7, S10, S11) into schedule because productivity is often measured with respect to time. Meanwhile, effort (S9) and estimating work size (S4, S10) are mapped into resources, as determining effort and estimating work level is part of the human resource management knowledge area in PMBOK. We found different variables and measures of quality. In mapping quality measures and variables, we used ISO/IEC (2010) as a reference to map the secondary studies. The variables that we found on quality are all related to software product quality, e.g., number of defects, size, etc. For simplicity, we refer to software product quality as quality in this paper. Table 10 summarizes the mapping of different variables and measures and project constraints.

We found that number of defects and defect density are commonly used to measure external quality. However, we found different variations of measures to indicate time, e.g., lead time, testing time, time between two subsequent releases, etc. We also found measures with different levels of granularity being used to measure internal quality, e.g., some secondary study mention a generic measure, like code quality, some specified more particular measures, like complexity, size, etc. Many secondary studies do not provide information on measures under investigation.

### 5.3. Quality assessment results

The quality scores for secondary studies are between 2–8, see Table 11. The maximum score that a secondary study could obtain is nine.

Five secondary studies scored three and below (S1, S5, S8, S12, S13). Most of the low-scoring studies are missing quality criteria for preventive steps to minimize bias (C5) and an actual synthesis of the found evidence (C7a and C7b). Criteria like minimizing bias, providing sufficient information on the included primary studies, synthesizing evidence, and evaluating the strength of evidence are

**Table 10**

Mapping of variables and measures to project constraints.

| Project Constraints | Variables                       | Measures                                                                                                                                                                                                                                                                                                                                                                     | Reference           |
|---------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Scope<br>Quality    | Requirements                    | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S12                 |
|                     | External Quality                | number of defects/bugs/ trouble reports, defects per KLOC/defect density, number/percentage of acceptance/ black-box/ external tests passed, number/percentage of unit tests passed, quality mark given by client                                                                                                                                                            | S11                 |
|                     |                                 | Defect count, defect density, mean time assertion per method, total assertions (passed/failed), number of test cases passed                                                                                                                                                                                                                                                  | S7                  |
|                     |                                 | number of passed acceptance tests or the total number of defects or number of defects/KLOC (defect density). number of defects found before release or defects reported by customers                                                                                                                                                                                         | S10                 |
|                     |                                 | Number of defects                                                                                                                                                                                                                                                                                                                                                            | S3, S8, S9          |
|                     |                                 | Robustness                                                                                                                                                                                                                                                                                                                                                                   | S7                  |
|                     | Customer satisfaction           | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S1, S4              |
|                     | Internal Quality                | Condition coverage, Branch coverage, Nested block depth, Cyclomatic complexity, Number of parameters, Coupling between objects, Information flow, Weighted class per method, Lack of cohesion metrics, Mutation score indicator                                                                                                                                              | S7                  |
|                     |                                 | Complexity, size                                                                                                                                                                                                                                                                                                                                                             | S2                  |
|                     |                                 | Code size, Cyclomatic complexity, Code reuse, Coupling and cohesion                                                                                                                                                                                                                                                                                                          | S10                 |
|                     | Code quality                    | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S2, S4, S5, S6, S12 |
|                     | Reusability software components | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S13                 |
| Schedule            | Time                            | Release cycles, Time between two subsequent releases                                                                                                                                                                                                                                                                                                                         | S8, S9              |
|                     |                                 | Mean-time per assertion, Mean time, Total/overall time, Average hours worked, Time taken in person hours, Time taken on testing as a percentage of total time, Time spent on coding as a percentage of total time, Total person hours                                                                                                                                        | S7                  |
|                     |                                 | Development time                                                                                                                                                                                                                                                                                                                                                             | S10                 |
|                     |                                 | Lead time                                                                                                                                                                                                                                                                                                                                                                    | S1                  |
|                     |                                 | Development time, feedback time                                                                                                                                                                                                                                                                                                                                              | S2                  |
|                     | Productivity                    | Testing time                                                                                                                                                                                                                                                                                                                                                                 | S12                 |
|                     |                                 | Output/unit effort, Total time to complete a story card, New lines of code/ development effort, LOC/person hour, LOC/development effort, User stories/ person hour                                                                                                                                                                                                           | S7                  |
|                     |                                 | Development time/person hours spent/task time, total LOC divided by total effort, or the number of LOC per hour, total noncommented LOC, number of delivered stories per unit effort (or implemented user stories per hour), delivered noncommented LOC per unit development effort (or effort per ideal programming hour), and hours per feature/development effort per LOC | S11                 |
|                     |                                 | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S10                 |
|                     |                                 | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S2                  |
| Budget              | Cost                            | Development cost                                                                                                                                                                                                                                                                                                                                                             | S10                 |
| Resource            | Work size                       | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S4, S10             |
|                     | Effort                          | Amount of Rework                                                                                                                                                                                                                                                                                                                                                             | S9                  |
|                     |                                 | Testing effort, development effort                                                                                                                                                                                                                                                                                                                                           | S10                 |
| Risk                | Adherence                       | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S7                  |
|                     | Developers' Opinion             | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S7                  |
|                     | Technical & Market Risk         | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S8                  |
| Communication       | Customer collaboration          | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S4                  |
|                     | Team work                       | Not provided                                                                                                                                                                                                                                                                                                                                                                 | S1, S4, S10         |

**Table 11**

Quality assessment.

| ID         | C1<br>RQ   | C2<br>Search | C3<br>Incl./Excl. | C4<br>QA | C5<br>Bias | C6<br>Detail | C7a<br>Synthesis | C7b<br>SoE | C8<br>Concl. | Sum |
|------------|------------|--------------|-------------------|----------|------------|--------------|------------------|------------|--------------|-----|
| S1         | 0,5        | 0,5          | 0,5               | 0        | 0          | 0,5          | 0                | 0          | 1            | 3   |
| S2         | 0,5        | 0,5          | 1                 | 0        | 0          | 1            | 1                | 0          | 1            | 5   |
| S3         | 0,5        | 1            | 1                 | 0        | 0          | 0,5          | 0,5              | 0          | 1            | 4,5 |
| S4         | 0,5        | 1            | 1                 | 1        | 1          | 1            | 0,5              | 1          | 1            | 8   |
| S5         | 0,5        | 0,5          | 0                 | 1        | 0          | 0            | 0                | 0          | 0            | 2   |
| S6         | 1          | 0,5          | 1                 | 0        | 0          | 0,5          | 0,5              | 0          | 1            | 4,5 |
| S7         | 0,5        | 0,5          | 1                 | 1        | 1          | 1            | 1                | 1          | 1            | 8   |
| S8         | 0,5        | 0,5          | 1                 | 0        | 0          | 0            | 0                | 0          | 1            | 3   |
| S9         | 0,5        | 0,5          | 1                 | 0        | 1          | 0,5          | 0,5              | 0          | 1            | 5   |
| S10        | 0,5        | 1            | 0,5               | 1        | 0          | 1            | 0,5              | 0          | 1            | 5,5 |
| S11        | 0          | 0,5          | 0                 | 0,5      | 0          | 1            | 1                | 1          | 1            | 5   |
| S12        | 0,5        | 0,5          | 1                 | 0,5      | 0          | 0,5          | 0                | 0          | 0            | 3   |
| S13        | 0,5        | 0,5          | 1                 | 0        | 0          | 0            | 0,5              | 0          | 0            | 2,5 |
| <b>Sum</b> | <b>6,5</b> | <b>8</b>     | <b>10</b>         | <b>5</b> | <b>3</b>   | <b>7</b>     | <b>6</b>         | <b>3</b>   | <b>10</b>    |     |



Fig. 4. Clusters of Secondary Studies Based on Quality Scores.

Table 12

Codes of contexts and research methods and their definitions.

| Context  | Research method |                  |    |
|----------|-----------------|------------------|----|
| Academic | A               | Case Study       | C  |
| Industry | I               | Experiment       | E  |
|          |                 | Mixed Study      | MS |
|          |                 | Multi-case Study | MC |
|          |                 | Simulation Study | SI |
|          |                 | Survey           | S  |

often missed or only partially addressed in most of the secondary studies.

Most secondary studies have clear research questions. However most of them are not formulated with respect to population, intervention, comparator, outcome and study design (PICOS). Most secondary studies often refer to another publication pertaining to their quality criteria and do not present the results of their quality assessments. Regarding minimizing bias, few secondary studies report inter-rater agreement (e.g., Kappa coefficient) or piloting of the review. Lastly, strengths of evidence are often not considered.

When we look at the quality scores of the secondary studies, we can see three different clusters, they are: (1) low quality, (2) medium quality, (3) high quality, see Fig. 4. The clusters of quality will be used in the analysis of the secondary studies in Section 5.4.

#### 5.4. Empirical support for the impacts of agile and lean practices

In the following subsections, we use the quality score clusters from Section 5.3 to map the impacts of Agile and Lean practices on project constraints according to the secondary studies summarized in Table 9. It should be noted that the following subsections are not a continuation nor part of the quality assessment. We use a '+' sign to indicate that an Agile and Lean practice has a positive impact on a project constraint. A '0' indicates that the Agile and Lean practice has no impact and a '-' indicates that it has a negative impact on a project constraint.

For each Agile and Lean practice, we present the number and types of primary studies, e.g., experiment, case study, or survey, that support a particular impact relationship ('+', '0', or '-'). If the type of a primary study is unclear, we note the type as "study". If the secondary study did not report on a particular impact relationship of an Agile and Lean practice and a project constraint, we leave the table cell blank.

To simplify the presentation of the tables in Sections 5.4.1–5.4.3, we use codes that represent the research methodologies and contexts of the primary studies, e.g., IC refers to an industry case study (see Table 12).

##### 5.4.1. High quality secondary studies

In this subsection, we report the impacts of Agile and Lean practices on different project constraints as reported in high quality secondary studies (S4, S7). There are four project constraints that are reported in high quality secondary studies, and these are, quality, schedule, resource, and communication.

In S7, the primary studies were categorized along their rigour and relevance according to Ivarsson and Gorschek (2011). Rigour captures how well a study is designed and executed, in particular the extent to which these aspects are reported and can be evaluated. Relevance captures the potential impact of a study on academia and industry. A large scale case study in an industrial context would for example be more industry relevant than an artificial academic experiment with students as subjects. In the following subsections we will refer to rigour and relevance as described above.

**Quality.** Different measures of quality are reported, they include, external quality, complexity, code size, etc. Empirical studies of varying rigour and relevance in S7 are largely in agreement that TDD has a positive impact on external quality. A small number of empirical studies with low relevance suggest that TDD has no impact on external quality.

Empirical studies with low relevance in S7 are in agreement that TDD has no impact on internal quality. Meanwhile two empirical studies involving students show that TDD has a positive impact on internal quality. We also observed that empirical studies with low relevance in S7 do not show agreement with respect to the impact of TDD on number of test and code size. For the individual mapping of other Agile and Lean practices and their impacts on quality as reported in high quality secondary studies, see Table 13.

**Schedule.** There are different measures associated with schedule such as, time, productivity, and progress tracking. Empirical studies, of high and low relevance, reported in S7 do not show agreements regarding the impacts of TDD on time. Empirical studies show that TDD could have a positive, negative, or no impact on time. A similar trend is also shown for the impacts of TDD on productivity. For the individual mapping of other Agile and Lean practices and their impacts on schedule as reported in high quality secondary studies, see Table 14.

**Resource.** Table 15 shows the impact of Agile and Lean practices on resource as reported in high quality secondary studies. The measures associated with resources include developers' opinion, conformance, and work estimation.

Empirical studies in S7 show that TDD has a positive impact on developers' opinion. The impact is supported by two surveys with high rigour and high relevance. One industry experiments with high rigour and low relevance reports that adherence to TDD process has a positive impact. In S4, one academic case study indicates that planning game has a positive impact on work estimation.

**Communication.** Table 16 shows the impact of Agile and Lean practices on communication. The measures associated with communication include customer collaboration and team work.

Empirical studies reported in S4 show positive and negative impacts of on-site customer on customer collaboration. One industry case study reports that planning game has a positive impact on team work (S4).

**Summary:** There are two secondary studies in the high quality cluster. S4 and S7 do not have overlapping primary studies (see Section 5.1 for primary studies overlap). Both secondary studies use SLR as a review method and (Kitchenham and Charters, 2007) as a guideline (see Section 4.1.1 for secondary study guidelines).

Between the two secondary studies, S7 provides more detailed information on the included primary studies than S4. S7 is the only secondary study that classifies its primary studies based on their rigour and relevance. The details provided by S7 could be influenced by its aim to examine impacts of TDD based on the primary studies' rigour and relevance. Meanwhile S4 is aimed to identify factors in Agile software development (see Section 4.1.2 for secondary studies' aims). From S7, we also observed that experiment studies, both in industry and academia, have a tendency to report that TDD has no impact on different measures of a project constraint.



**Table 13**  
Impact of agile practices on quality in high quality secondary studies.

| Practice         | Measure          | Ref. | Quality                  |                           |                   |
|------------------|------------------|------|--------------------------|---------------------------|-------------------|
|                  |                  |      | +                        | 0                         | -                 |
| Pair programming | External quality | S4   | 1 AC                     |                           |                   |
| Short iteration  | External quality | S4   | 1 IMC                    |                           |                   |
| TDD              | External quality | S7   | 6 ICs, 1 S <sup>a</sup>  |                           |                   |
|                  |                  | S7   | 4 ICs <sup>b</sup>       |                           |                   |
|                  |                  | S7   | 2 AEs, 1 IE <sup>c</sup> | 6 AEs <sup>c</sup>        | 1 AE <sup>c</sup> |
|                  |                  | S7   | 1 S <sup>d</sup>         | 1 AE <sup>d</sup>         |                   |
|                  |                  | S7   |                          |                           |                   |
|                  | Internal quality | S4   | 1 AC                     |                           |                   |
|                  |                  | S7   | 1 AE <sup>c</sup>        | 2 IEs, 8 AEs <sup>c</sup> |                   |
|                  |                  | S7   |                          | 1 IE, 2 AEs <sup>d</sup>  |                   |
|                  | Complexity       | S7   | 2 ICs <sup>a</sup>       |                           |                   |
|                  | Robustness       | S7   |                          | 1 IE <sup>d</sup>         |                   |
|                  | No of test       | S7   | 1 IE <sup>c</sup>        | 2 AEs, 1 IE <sup>c</sup>  | 1 AE <sup>c</sup> |
|                  |                  | S7   |                          | 1 IE, 1 AE <sup>d</sup>   |                   |
|                  | Code size        | S7   |                          | 2 AEs <sup>c</sup>        | 1 IE <sup>c</sup> |
|                  |                  | S7   | 1 IE <sup>d</sup>        |                           |                   |

<sup>a</sup> Studies with high rigour and high relevance.

<sup>b</sup> Studies with low rigour and high relevance.

<sup>c</sup> Studies with high rigour and low relevance.

<sup>d</sup> Studies with low rigour and low relevance.

**Table 14**  
Impact of agile practices on schedule in high quality secondary studies.

| Practice         | Measure           | Ref. | Schedule                |                           |                    |
|------------------|-------------------|------|-------------------------|---------------------------|--------------------|
|                  |                   |      | +                       | 0                         | -                  |
| TDD              | Time              | S7   |                         |                           | 2 ICs <sup>b</sup> |
|                  |                   | S7   | 4 AEs <sup>c</sup>      | 1 AE, 1 IE <sup>c</sup>   |                    |
|                  |                   | S7   | 1 IC, 1 AE <sup>d</sup> |                           |                    |
|                  | Productivity      | S7   |                         | 1 IC <sup>a</sup>         | 2 ICs <sup>a</sup> |
|                  |                   | S7   |                         |                           | 1 IC <sup>b</sup>  |
|                  |                   | S7   | 1 AE <sup>c</sup>       | 2 IEs, 8 AEs <sup>c</sup> |                    |
| Pair programming | Time              | S4   | 1 S                     |                           | 1 S <sup>d</sup>   |
| Stand up meeting | Progress tracking | S4   | 1 IC, 1 IMC             |                           | 1 AC               |

<sup>a</sup> Studies with high rigour and high relevance.

<sup>b</sup> Studies with low rigour and high relevance.

<sup>c</sup> Studies with high rigour and low relevance.

<sup>d</sup> Studies with low rigour and low relevance.

**Table 15**  
Impact of agile practices on resource in high quality secondary studies.

| Practice      | Measure             | Ref. | Resource                |                    |   |
|---------------|---------------------|------|-------------------------|--------------------|---|
|               |                     |      | +                       | 0                  | - |
| TDD           | Developers' opinion | S7   | 2 Surveys <sup>a</sup>  |                    |   |
|               |                     | S7   | 1 IC, 1 AE <sup>c</sup> | 4 AEs <sup>c</sup> |   |
|               |                     | S7   | 2 AEs <sup>d</sup>      |                    |   |
|               | Adherence           | S7   | 1 IE <sup>c</sup>       |                    |   |
| Planning game | Work estimation     | S4   | 1 AC                    |                    |   |

<sup>a</sup> Studies with high rigour and high relevance

<sup>b</sup> Studies with low rigour and high relevance

<sup>c</sup> Studies with high rigour and low relevance

<sup>d</sup> Studies with low rigour and low relevance

**Table 16**  
Impact of agile practices on communication in high quality secondary studies.

| Practice         | Measure                | Ref. | Communication |   |        |
|------------------|------------------------|------|---------------|---|--------|
|                  |                        |      | +             | 0 | -      |
| On-site customer | Customer collaboration | S4   | 1 IC          |   | 2 IMCs |
| Planning game    | Team work              | S4   | 1 IC          |   |        |

We found a large number of primary studies that report on positive impacts but we cannot dismiss the smaller number of studies that reported on negative or no impacts. The classification made in S7 based on rigour and relevance is helpful to deduce the im-

pact of an Agile and Lean practice. However, we still cannot make strong assertions on the impact of an Agile and Lean practice. Primary studies with high rigour and high relevance are more trustworthy. However, we cannot dismiss studies of low rigour and low relevance. The heterogeneity of the data made aggregating findings from the two secondary studies difficult.

#### 5.4.2. Medium quality secondary studies

In this section, we report the impacts of Agile and Lean practices on different project constraints as reported in medium quality secondary studies (S2, S3, S6, S9, S10, S11). There are four project constraints that were reported in medium quality secondary studies, and these are: quality, schedule, budget, resource, and communication.

**Quality.** The measures associated with quality include external quality, complexity, cohesion, code reusability, etc. Empirical studies in S3, S10, and S11 do not show agreement regarding the impacts of TDD on external quality. We found a number of empirical studies with different research methods and contexts that indicate TDD has a positive impact on external quality. However, a number of experiments also indicate that TDD could have no impact as well as negative impact on external quality.

We also observed that empirical studies in S2 and S10 are largely in agreement with respect to the positive impact of TDD on complexity. Only one empirical study in S2 that suggests TDD has a negative impact on complexity. For the individual mapping

**Table 17**  
Impact of agile practices on quality in medium quality secondary studies.

| Practice           | Measure          | Ref. | Quality                   |         |                                                |
|--------------------|------------------|------|---------------------------|---------|------------------------------------------------|
|                    |                  |      | +                         | 0       | -                                              |
| Refactoring<br>TDD | Quality          | S10  | 1 AC, 1 IC                |         |                                                |
|                    | External quality | S3   | 1 study                   |         |                                                |
|                    |                  | S10  | 5 AE, 1 Survey, 1 IE      | 2 AE    |                                                |
|                    |                  | S11  | 8 IEs                     | 18 AEs  | 7 experiments comparing to iterative test last |
|                    | Complexity       | S2   | 1 (IE+AE+IC), 3 ICs, 1 AE |         | 1 IC                                           |
|                    |                  | S10  | 2 ICs                     |         |                                                |
|                    | Code quality     | S2   | 9 ICs, 2 AEs, 1 AC, 1 IE  | 2 AEs   | 1 AE                                           |
|                    |                  | S6   | 1 SI                      |         |                                                |
|                    | Code size        | S2   | 1 (IE+AE+IC)              |         | 1 AE                                           |
|                    | Cohesion         | S10  |                           | 1 AE+IE |                                                |
| Pair programming   | Code reusability | S10  | 1 AE                      |         |                                                |
|                    | Code quality     | S10  | 5 AEs, 1 IE               |         |                                                |
|                    | External quality | S10  | 1 S, 1 AE                 | 1 AE    |                                                |
| VSM                | External Quality | S9   | 2 studies                 |         |                                                |

**Table 18**  
Impact of agile practices on schedule in medium quality secondary studies.

| Practice                   | Measure          | Ref. | Schedule     |                                                |                          |
|----------------------------|------------------|------|--------------|------------------------------------------------|--------------------------|
|                            |                  |      | +            | 0                                              | -                        |
| Refactoring<br>TDD         | Productivity     | S10  | 3 ICs, 1 AMS |                                                |                          |
|                            | Development time | S2   | 3 AEs, 2 ICs | 1 IC, 1 AE                                     | 6 ICs, 1 IE, 3 AEs, 1 AC |
|                            |                  | S10  |              |                                                | 2 IC                     |
|                            | Feedback time    | S2   | 3 ICs, 1 S   |                                                |                          |
|                            | Productivity     | S10  | 2 AEs        | 1 IC                                           | 1 IC                     |
| Pair programming           | Productivity     | S11  | 18 AEs       | 7 experiments comparing to iterative test last | 8 IEs                    |
|                            |                  | S10  |              | 1 IC, 1 IE                                     |                          |
|                            | Schedule         | S10  |              | 1 IE, 1 S, 1 AC, 2 AEs                         |                          |
| Internal/ external release | Release cycle    | S9   | 1 study      |                                                |                          |

**Table 19**  
Impact of agile practices on cost in medium quality secondary studies.

| Practice         | Measure          | Ref. | Cost |             |   |
|------------------|------------------|------|------|-------------|---|
|                  |                  |      | +    | 0           | - |
| TDD              | Cost             | S2   | 1 IC |             |   |
| Pair programming | Development cost | S10  | 1 IC |             |   |
|                  | Cost             | S10  |      | 2 AEs, 1 AC |   |

of other Agile and Lean practices and their impacts on quality as reported in medium quality secondary studies, see [Table 17](#).

**Schedule.** The measures associated with schedule include productivity, time, and release cycle. Empirical studies in S2 and S10 are largely in agreement that TDD has a negative impact on development time. However, S2 also report a number of empirical studies that show contradictory results. Inconsistent impacts of TDD on productivity are also shown in the empirical studies reported in S10 and S11.

Meanwhile empirical studies in S10 show that pair programming has no impact on both productivity and schedule. For the individual mapping of other Agile and Lean practices and their impacts on schedule as reported in medium quality secondary studies, see [Table 18](#).

**Budget.** The empirical studies on the impacts of Agile and Lean practices on budget reported in medium quality secondary study is rather scarce. [Table 19](#) shows the impacts of Agile and Lean practices on budget as reported in medium quality secondary studies. The measures associated with budget include cost and development cost.

Empirical studies in S2 and S10 suggest that TDD has a positive impact on cost. S10 reports two academic experiments and one academic case study that suggest pair programming has a negative impact on cost.

**Table 20**  
Impact of agile practices on resource in medium quality secondary studies.

| Practice         | Measure              | Ref. | Resource |   |       |
|------------------|----------------------|------|----------|---|-------|
|                  |                      |      | +        | 0 | -     |
| TDD              | Testing effort       | S10  |          |   | 1 AE  |
|                  | Development effort   | S10  | 1 AE     |   |       |
|                  | Adherence            | S2   | 1 S      |   | 3 ICs |
| Pair programming | Effort               | S10  |          |   | 1 IE  |
| Planning game    | Work size estimation | S10  | 1 AC     |   |       |
| VSM              | Rework effort        | S9   | 1 study  |   |       |

**Table 21**  
Impact of agile practices on communication in medium quality secondary studies.

| Practice         | Measure  | Ref. | Communication          |   |   |
|------------------|----------|------|------------------------|---|---|
|                  |          |      | +                      | 0 | - |
| Pair programming | Teamwork | S10  | 3 AEs, 1 S, 1 AC, 1 IE |   |   |

### Resource

The empirical studies on the impacts of Agile and Lean practices on resources reported in medium quality secondary study is also scarce. We found only one empirical study for each impact of Agile and Lean practices on resources.

One academic experiment reports that TDD has a negative impact on testing effort. However, another academic experiment reports that TDD has a positive impact on development effort (S10). For the individual mapping of other Agile and Lean practices and their impacts on resources as reported in medium quality secondary studies, see [Table 20](#).

**Communication.** One medium quality secondary study reports the impact of an Agile practice on communication (see [Table 21](#)).

S10 reports three academic experiments, one survey, one academic experiment, and one industry experiment that indicate pair programming has a positive impact on teamwork.

**Summary:** None of the primary studies in the medium quality cluster were assessed based on rigour and relevance. Research methods and contexts of the primary studies are traceable for most of the primary studies. However, making conclusive remarks regarding the impact of Agile and Lean practices was difficult. Although we found a number of industrial empirical studies that report on a positive impacts, we cannot dismiss studies in academic settings with contrary results.

From the medium quality cluster, S2 and S10 have small number of overlapping primary studies. Both S2 and S10 examine the impact of TDD on complexity, development time, and cost. However, the primary studies included in S2 and S10 are not in agreement regarding the impacts of TDD on development time and cost. S2 report a positive impact and a negative impact of TDD on complexity. S10 only report a positive impact of TDD on complexity. However, S2 and S10 are in agreement with regards to a positive impact of TDD on cost.

We did not find other secondary studies in the medium quality cluster that have overlapping primary studies. However, S2 and S7 have many overlapping primary studies, see [Section 5.1](#). Both S2 and S7 examine the impact of TDD on the code quality, code size, complexity, time and adherence. However, the primary studies in S2 and S7 are not in agreement regarding the impacts of TDD on the aforementioned measures.

S2, S7, and S11 are secondary studies that focus on examining TDD. Except S7, S2 and S11 are in medium quality cluster. When we examined the impacts of TDD as indicated the primary studies (in S2, S7, and S11), we did not find agreements. However, when we looked at the conclusions made in S2, S7, and S11, we found similarities. Although S2, S7, and S11 use slightly different measures to assess quality, these secondary studies conclude that TDD has a positive impact on improving quality. S7 and S11 both conclude that the impact of TDD on productivity is inconclusive, however more likely to be negative.

From [Table 17](#), we observed that experiment studies conducted with students tend to report that Agile and Lean practices has no impact on quality measures. When we examined the secondary study methods, with the exception of S3, S6, and S9, the medium quality secondary studies used SLR as a method and ([Kitchenham and Charters, 2007](#)) as a guideline. S3 and S9 are both SM studies. S3 used the guideline from ([Petersen et al., 2008](#)), meanwhile S9 referred to ([Kitchenham and Charters, 2007](#)) as a guideline. S6 modified the steps from ([Kitchenham and Charters, 2007](#)) and referred the method as quasi-SLR.

When we examined the aims of the secondary studies in the medium quality cluster, S2, S10, and S11 are aimed to find association. It is evident from [Table 17 – 21](#), S2, S10, and S11 examine the associations of different Agile and Lean practices and their impacts of different project constraints. S6 is aimed to identify factors, meanwhile S3 and S9 are aimed to classify primary studies. Perhaps since S3, S6 and S9 are not aimed to find association, they only report positive impacts of Agile and Lean practices.

#### 5.4.3. Low quality secondary studies

In this section, we report the impacts of Agile and Lean practices on different project constraints as reported in low quality secondary studies (S1, S5, S8, S12, S13). There are five project constraints that are reported in low quality secondary studies, they are, scope, quality, schedule, risk, and communication.

**Scope.** We identified only one secondary study in the low quality cluster that reports the impact of Agile practices on scope (see [Table 22](#)). In S12, one case study and two studies suggest that TDD has a positive impact on requirements specification.

**Table 22**

Impact of agile practices on scope in low quality secondary studies.

| Practice | Measure      | Ref. | Scope          |   |   |
|----------|--------------|------|----------------|---|---|
|          |              |      | +              | 0 | - |
| TDD      | Requirements | S12  | 1 C, 2 studies |   |   |

**Quality.** Studies reported in S5 and S12 are in agreement that TDD has a positive impact of code quality. In S13, two studies suggest that TDD has a positive impact on component reusability. The contexts of the primary studies are not reported by the secondary studies. For the individual mapping of other Agile and Lean practices and their impacts on quality as reported in low quality secondary studies, see [Table 23](#).

**Schedule.** All primary studies included in low quality secondary studies indicate that Agile and Lean practices have positive impacts on different measures of schedule, see [Table 24](#). None of the secondary studies report the research methods and contexts of the primary studies.

**Risk.** We identified only one secondary study in the low quality cluster that reports the impact of Agile practices on risk (see [Table 25](#)). One study suggests that incremental deliveries has a positive impact on managing technical and market risks (S8).

**Communication.** All primary studies included in low quality secondary studies indicate that Agile and Lean practices have positive impacts on different measures of communication, see [Table 26](#). None of the secondary studies report the research methods and contexts of the primary studies.

**Summary:** All secondary studies in the low quality clusters do not mention the contexts of the primary study. With the exception of S12, most secondary studies also do not report the research methodology of the primary studies. We also observed that there is a tendency of bias from the low quality secondary studies. No secondary studies in the low quality cluster report negative impacts of Agile and Lean practices on the project constraints.

None of the secondary studies in the low quality cluster are aimed at finding association. With the exception of S1, most secondary studies are aimed at identifying factors. Meanwhile the aim of S1 is to survey the literature. The aims of the secondary studies could contribute the tendency of bias in the low quality cluster. Since the secondary studies are not aimed to examine the impacts of Agile and Lean practices.

When examining the overlapping of primary studies, S1 and S8 have one overlapping primary study. However, S1 and S8 do not examine the same Agile practices nor measures of project constraints.

With the exception of S8, most secondary studies in low quality cluster used SLR as a secondary study method with the guideline from ([Kitchenham and Charters, 2007](#)). In S8, the authors mentioned that they did not perform the complete SLR steps as mentioned in ([Kitchenham and Charters, 2007](#)), and referred to their secondary study method as semi-SLR.

#### 5.4.4. General summary on the impacts of agile and lean practices

We identified a large number of primary studies that study the impact of TDD on external quality. Many of those primary studies indicate that TDD has a positive impact on external quality. Unfortunately, we are unable to make any inference on the remaining Agile and Lean practices, due to inefficient data. For example, the positive impact of planning game on team work, incremental delivery on test predictability, and Kanban pull systems on work quality are only reported in one primary study.

**Table 23**  
Impact of agile practices on quality in low quality secondary studies.

| Practice               | Measure                        | Ref. | Quality                           |           |   |
|------------------------|--------------------------------|------|-----------------------------------|-----------|---|
|                        |                                |      | +                                 | 0         | - |
| Kanban pull systems    | Work quality                   | S1   | 1 study                           |           |   |
| Adaptive Planning      | Customer satisfaction          | S1   | 5 studies                         |           |   |
| TDD                    | Code quality                   | S5   | 2 studies in open source projects |           |   |
|                        |                                | S12  | 2 Cs, 1 MC, 2 studies             |           |   |
|                        | Software component reusability | S13  | 2 studies                         |           |   |
| Incremental deliveries | External quality               | S8   | 4 studies                         | 2 studies |   |
|                        | Test predictability            | S8   | 1 study                           |           |   |

**Table 24**  
Impact of agile practices on schedule in low quality secondary studies.

| Practice               | Measure       | Ref. | Schedule  |   |   |
|------------------------|---------------|------|-----------|---|---|
|                        |               |      | +         | 0 | - |
| Kanban pull systems    | Lead time     | S1   | 1 study   |   |   |
| Adaptive planning      | Efficiency    | S1   | 2 studies |   |   |
| Team chooses own task  | Time          | S1   | 1 study   |   |   |
| Incremental deliveries | Release cycle | S8   | 3 studies |   |   |
| TDD                    | Testing time  | S12  | 1 study   |   |   |

**Table 25**  
Impact of agile practices on risk in low quality secondary studies.

| Practice             | Measure                   | Ref. | Risk    |   |   |
|----------------------|---------------------------|------|---------|---|---|
|                      |                           |      | +       | 0 | - |
| Incremental delivery | Technical and market risk | S8   | 1 study |   |   |

**Table 26**  
Impact of agile practices on communication in low quality secondary studies.

| Practice               | Measure                | Ref. | Communication |   |   |
|------------------------|------------------------|------|---------------|---|---|
|                        |                        |      | +             | 0 | - |
| Adaptive planning      | Team work              | S1   | 5 studies     |   |   |
| Incremental deliveries | Customer collaboration | S8   | 1 study       |   |   |

## 6. Discussion

In this tertiary study, we reviewed 41 secondary studies in Agile and Lean software development. Thirteen out of 41 secondary studies reported the impacts of Agile and Lean practices on various measures of project constraints.

### 6.1. Reflections on secondary studies in agile and lean

Our tertiary study revealed a large number of secondary studies in Agile and Lean software development. However, many of them do not report clear review processes. Therefore, we only included 41 secondary studies after full text review.

We also found that most secondary studies are aimed at identifying factors in Agile and Lean software development such as, challenges or benefits of practices. This result is similar to the tertiary study by Verner et al. (2014), where they found that most SLRs in GSD produced summary and mappings of the literature. In our tertiary study, we identified four secondary studies that specifically report the impacts of Agile and Lean practices.

The guideline from Centre for Reviews and Dissemination (CRD) suggested that an SLR is aimed to evaluate and summarize available evidence and make them more accessible to decision makers (Centre for Reviews and Dissemination, 2009). However, the results from our quality assessment process shows that very few secondary studies performed quality evaluation of primary stud-

ies. This is inline with findings reported by Kitchenham et al. (2010) from a review of SLRs in software engineering.

One of the goals of EBSE is to provide a means for practitioners in making informed decisions about technology adoption (Kitchenham et al., 2004). Our tertiary study also revealed that very few secondary studies performed synthesis of the evidence. The lack of evidence synthesis hinders practitioners to make informed decisions pertaining to adoption of Agile and Lean practices. Without synthesized evidence, practitioners cannot determine whether adopting Agile and Lean practices would be beneficial for them. The results of our tertiary study show that most secondary studies mentioned that they performed SLRs. Yet, most of the SLRs are aimed at identifying factors, i.e., used Agile and Lean practice, benefits, and limitations, instead of synthesizing evidence pertaining to the impacts of Agile and Lean practices.

The lack of overlap of primary studies indicates the lack of a common goal among researchers, which is one of the goals of EBSE. It indicates that researchers are not building on top of each others' work. Instead individual researchers, or research groups are conducting their own search process. The lack of a common goal among researchers is also reflected on the breadth of topics being investigated in the secondary studies of Agile and Lean software development.

Our tertiary study also revealed "simplified" versions of an SLR, they are, quasi-SLR (S6) and semi-SLR (S8). Both in S6 and S8, the authors mentioned that they did not perform a number of steps that are mentioned in the SLR guideline. Therefore their secondary study would not fit a "complete" SLR. Kitchenham et al.'s reported that one of the key issues in performing SLR is the time spent to perform the steps (Kitchenham et al., 2009). Our study revealed a number of alternatives that have been performed to simplify the SLR steps.

### 6.2. Reflections on the impacts of agile and lean practices

From 41 secondary studies only 13 provided discussions on the impact of Agile and Lean practices. Out of 26 practices mentioned in Petersen (2011), we found 13 practices that were mentioned in the secondary studies. Four out of 13 secondary studies are dedicated to specific practices, i.e., TDD (S2, S7, S11), and incremental deliveries (S8). TDD is the most studied Agile practice. TDD is mentioned in 10 secondary studies. Hanssen et al. (2011) performed a tertiary on GSD and identified a number of Agile practices that are used in GSD contexts, such as, continuous integration, pair programming, retrospective, and daily stand up meetings. Our tertiary study also identified the same practices, with the exception of continuous integration. Our study shows there are many Agile and Lean practices are still under-studied, e.g., refactoring, stand-up meeting, etc., see Table 9. Our findings do not imply that the rest of Agile and Lean practices are not evaluated at all. The reason why some practices are under-studied could be attributed to the decline in the adoption of some Agile and Lean practices, such as,



TDD, pair programming, and continuous integration (Murphy et al., 2013; Solinski and Petersen, 2014).

We identified a number of measures that are associated with a project constraint. For instance, number of defects is the most commonly used measure of quality. However, we did not find commonly used measures for other project constraints. For instance, we identified a number of measures that are associated with schedule, e.g., time between two releases, lead time, development time, testing time, etc. The measures used in the secondary studies also have varying granularity levels. For instance, some secondary studies used more specific measures like complexity or code size. However other secondary studies used a generic measure like code quality. Our finding suggests that currently there is no agreed measures to assess the impacts of Agile and Lean practices on project constraints. The variations of measures and their level of granularity made the aggregation of findings more difficult. Similar observation is shared in S11 (Rafique and Mišić, 2013), where no common metrics were used to measure internal quality. As a consequence Rafique and Mišić (2013) were not able to consider internal quality in the analysis of TDD's impact on quality.

In this tertiary study, we also examined the overlap of primary studies between the secondary studies. We found little overlap between the secondary studies, even those with similar topics. The largest overlap of primary studies is between S2 and S7 with 21 overlapping primary studies. Both S2 and S7 are aimed at finding the impacts of TDD, particularly on quality and schedule. There are a number of measures that are examined in S2 which are also examined in S7, such as, code size, complexity, adherence, and time. Despite similar aims and large overlap of primary studies, the primary studies in S2 and S7 are not in agreement with respect to the impacts of TDD on code size, complexity, adherence, and time. Our tertiary study shows that two secondary studies with a similar topic may not yield the same results. Our finding is similar to the tertiary study conducted by Wohlin et al. (2013), where the results of a secondary study may not be reproducible.

Three secondary studies, S2, S7, and S11, are dedicated to examining the impacts of TDD. The primary studies of S2, S7, and S11 show contradictory results regarding the impacts of TDD. However, when we examined the conclusions in S2, S7, and S11, we identified similarities. S2, S7 and S11 indicate that TDD has a positive impact on quality, despite the lack of overlapping primary studies between S7 and S11. Meanwhile, S7 and S11 conclude that the impact of TDD on productivity is inconclusive, with a tendency towards negative impact. Our tertiary study shows that two secondary studies with a similar topic can have similar conclusions. Our finding is similar to the tertiary done by MacDonell et al. (2010) where two secondary studies with a common research question could yield consistent results.

Wohlin et al. (2013) and MacDonell et al. (2010) found contradicting results regarding the reliability and reproducibility of secondary studies' results. Our tertiary study finds similarities with both tertiary studies. When we examine the conclusions of the secondary studies with a common topic we identified commonalities. However, when we examine the finer details of the primary studies that are included in the secondary studies, we identified inconsistencies. Our tertiary study shows that there are different perspectives within a secondary study. Depending on the readers' perspectives, the outcomes of a secondary study may be more or less relevant. Some readers might be more interested in the general picture, some readers might be more interested in the detailed picture.

Given the high interest in TDD, it is important to bring up the issue of adherence to TDD process. S11 reports that often primary studies do not report the level of adherence to the description of TDD. S2 and S7 report that one industry experiment and one survey suggest that adherence to TDD process has a positive impact.

However, S2 also reports that three industry case studies suggest adherence to TDD has a negative impact. Our study indicates the importance of reporting the level of adherence in future TDD studies. Also, the importance of further examining the impacts of low adherence versus high adherence to TDD process.

When we examined the clusters of quality scores, we did not find a distinctive outcome from each cluster with respect to the impact of Agile and Lean practices. However, high and medium quality secondary studies provide sufficient details with regards to the primary studies' method and contexts. One secondary study in the high quality cluster (S7) provide additional information pertaining to the rigour and relevance of the primary studies. This information helped us to discern how and in which contexts the results were obtained.

From the high and medium quality secondary studies clusters we observed that experiment studies, especially with students, tend to show that an Agile and Lean practice has no impact on any of the project constraints measures. The result of our study shows that we need to examine experiment studies more closely, especially those that involve students. Our study indicates that we need to better understand why experiment studies tend to show no impact. We need to investigate if the trend is influenced by experiment design, the formulated hypotheses, or selected confidence level.

We also uncovered a trend of publication bias from the low quality secondary studies cluster. All secondary studies in the low quality cluster did not report on the negative impacts of Agile and Lean practices. However, the tendency of bias from secondary studies in the low quality cluster is influenced by the aims of the secondary studies. Secondary studies in the low quality cluster are primarily aimed to survey the literature and identify factors. Our study shows the possibility of publication bias could be influenced by the aims of the secondary studies themselves.

Our tertiary study reveals that there is a significant amount of data pertaining to TDD, especially on the impact of TDD on external quality. The secondary studies show that TDD has a positive impact on external quality. However, it is important to note that this inference is solely based on the number of primary studies. We are not able to make a strong assertion due to the heterogeneity of the data. For the most part we do not know the quality of the primary studies that report on the impact of TDD on external quality. Also, currently, there is no known approach to aggregate findings from different research methods, such as, case studies and experiments. Furthermore, currently there is no agreed method on how to value or weigh the results of studies with high rigour and relevance versus studies of low rigour and relevance. We also do not have an agreed method on how to value studies with practitioners versus studies with students.

We are unable to make similar inference for the other Agile and Lean practices and their impacts on project constraints. Most of them are only reported in one or two secondary studies. Furthermore, the number of primary studies that we identified from the secondary studies are not sufficient to make any remarks.

### 6.3. Implications for research

The first implication for research is our tertiary study shows the need to further investigate the benefits and trade-offs of Agile and Lean practices. Our tertiary study shows that an Agile practice, such as TDD can have both a positive impact on one project constraint and a negative impact on another project constraints. Our tertiary study reveals the need to put effort into investigating benefits and trade-offs of combining different Agile and Lean practices.

In this tertiary study, we uncovered the contradictory outcomes of Agile and Lean practices obtained from different secondary studies with varying degree of quality. The heterogeneity of the data

made it difficult to come to conclusive remarks pertaining to the impacts of Agile and Lean practices. The second implication for research from our tertiary study is the need to translate the knowledge obtained from this tertiary study, as well as existing secondary studies, to practitioners. This is known as knowledge translation (Kitchenham et al., 2015). As researchers we need to involve industry practitioners on how to better interpret and formulate good recommendations based on the results of secondary and tertiary studies. The mapping of Agile and Lean practices and their impacts on project constraints based on this tertiary study could be used as a starting point to initiate knowledge translation process.

Our tertiary study also shows that there are different perspectives to a secondary study. The third implication for research from our tertiary study for future secondary studies is to include "perspectives" in the implication for practice section. Practitioners and managers who read the secondary study will be able to navigate through different parts of the findings that are more relevant to them. For instance, managers might be more interested in the general findings. Meanwhile practitioners might be more interested in the detailed impacts on certain measures.

Our tertiary study also reveals gaps in Agile and Lean research which opens up an avenue for further research. There are still many Agile and Lean practices that have not been explored, like metaphors, coding standards, etc. We need to better understand why these practices are not investigated as intensively as the others. With new studies that show that some practices are declining in their adoption (Murphy et al., 2013; Solinski and Petersen, 2014), it is also worth investigating the reasons why practices, like TDD and pair programming, that appear to have a positive impact on quality are declining in their adoption.

#### 6.4. Implications for practice

Our tertiary study compiled a list of Agile and Lean practices and their impacts (positive, negative, or no impact) on various measures of project constraints. Although we did not provide suggestive propositions, practitioners can use our findings to help with assessing the suitability, as well as possible trade-offs, of adopting an Agile and Lean practice. For instance, a number of secondary studies report that TDD has a positive impact on external quality, but TDD might compromise productivity. If a practitioner wants to adopt TDD, they could expect improvement in external quality however, it might negatively affect productivity.

Both practitioners and managers can benefit from our tertiary study as it provides an overview of the impacts each Agile and Lean practice on different measures of project constraints. Practitioners can benefit from understanding better the impact of an Agile and Lean practice on different measures of a project constraint. Managers can benefit from getting an overview of the potential impacts that an Agile and Lean practice can have on a project constraint.

## 7. Conclusions

The aim of this tertiary study is to provide a consolidated view of findings pertaining to the impacts of Agile and Lean practices on project constraints. The aim was achieved by conducting a tertiary study in Agile and Lean software development. We performed an automated database search and found 122 secondary studies in Agile and Lean software development. From the 122, only 41 secondary studies passed the inclusion/exclusion criteria. Only 13 out of 41 secondary studies provided discussions pertaining to the impacts of the Agile and Lean practices.

We identified 13 Agile and Lean practices and their impacts on different project constraints. TDD and its impact on quality is the

most studied practice-impact association. Given the heterogeneity of the data we were unable to perform a synthesis. Instead we mapped the findings based on the quality scores of the secondary studies and the respective Agile and Lean practices and their impacts.

### 7.1. Revisit the research questions

In this subsection, we revisit the research questions that were formulated in this study.

*RQ1: What are the sets of primary studies that the secondary studies draw upon?* The secondary studies in Agile and Lean software development draw from a large set of primary studies. The overlap between the secondary studies is small. Most secondary studies do not have overlaps with each other. There are 420 primary studies that are included in 13 secondary studies that we included for detailed analysis. Only 50 out of 420 primary studies are overlapping. The main reason for overlapping primary studies is a similarity in topic, such as Agile/SPLE and TDD. However, the distinctions in the review designs of the secondary studies leads to the small number of overlap. For instance S7, and S11 investigate the impacts of TDD. However, S11 is focused on experiments in TDD, meanwhile S7 includes different types of empirical studies.

*RQ2: What are the scopes of the secondary studies in evaluating Agile and Lean software development?* We identified 13 Agile and Lean practices out of 26 that are mentioned in Petersen (2011), e.g., refactoring, pair programming, TDD, etc. The list of 13 practices can be found in Section 5.2.1. We also found the impacts of the 13 Agile and Lean practices on various measures of project constraints, such as, external quality, internal quality, time, etc. The list of measures that are used in the secondary studies can be found in Section 5.2.2.

*RQ3: What is known about the impacts of Agile and Lean practices?* We found that different Agile and Lean practices can have positive, negative, or no impacts on the different measures of project constraints. The impacts of each Agile and Lean practices can be found in Section 5.4.1, 5.4.2, and 5.4.2. We did not find strong agreements between the secondary studies, even the ones with a similar topic and large overlap of primary studies. When we examined the secondary studies based on their quality clusters, we also did not find strong distinctions between the cluster in terms of the impacts of Agile and Lean practice. However, we found that high and medium quality clusters provide sufficient information pertaining to the primary studies' contexts, rigour and relevance, and used empirical methods. Secondary studies in the low quality cluster do not provide any information regarding the contexts nor the empirical method of the primary studies. Secondary studies in low quality cluster also show a tendency of publication bias, because they do not report primary studies with negative impacts.

This tertiary study reveals the adverse impacts of Agile and Lean practices. We identified a significant number of primary studies that suggest that TDD has a positive impact on external quality. The heterogeneity of the data does not allow us to make a conclusive assertion. For the remaining Agile and Lean practices that we identified in this tertiary study, we are unable to make similar remarks due to insufficient data. However, we provide a consolidated view of the impacts of Agile and Lean practices and the number of primary studies that support them, including their rigour and relevance, research method, and contexts. For future work, we propose the following: (1) study the trade-offs associated with implementing a combination of Agile and Lean practices, (2) investigate the impacts of other Agile and Lean practices that have not been identified in this tertiary study, e.g., metaphors and user stories, coding standard, and continuous integration (3) investigate why some Agile and Lean practices are declining in adoption despite the high number of empirical studies that show their benefits.

**Table A.27**

Aims of secondary studies.

| Type                     | Ref | Aim                                                                                                                                                                                                                                                          |
|--------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classify Primary Studies | S3  | Collect evidence about how Agile Software Product Line (SPL) research is structured, synthesize current evidence on the integration of both approaches and identify further challenges for the integration of Agile methods and SPL.                         |
|                          | S9  | Identify and classify state of the art in large-scale software development influenced by Lean Product Development (LPD) approaches.                                                                                                                          |
|                          | S19 | Analyze agile practices in order to explore their industrial usage with respect to their distribution over different domains and processes from the perspective of software engineers.                                                                       |
|                          | S21 | Investigate the state of the art on how standardized model like Agile can work with Model Driven Development (MDD)                                                                                                                                           |
| Find association         | S24 | Identify relevant research and understand what the field of Agile-User Experience (UX) looks like at present.                                                                                                                                                |
|                          | S2  | Investigate potential factors that are limiting the industrial adoption of Test Driven Development (TDD).                                                                                                                                                    |
|                          | S7  | Investigate impact of TDD on different variables whilst taking two study quality dimension into account, namely rigor and relevance.                                                                                                                         |
|                          | S10 | Evaluate according to the ISO/IEC 12,207 and ISO/IEC 9126 standards, synthesize, and present, the empirical findings on quality in agile methods.                                                                                                            |
| Identify Factors         | S11 | Investigate the impact of TDD on external code quality and productivity.                                                                                                                                                                                     |
|                          | S26 | Gather and synthesize empirical evidence to provide convincing and illuminating support for software project managers who need to make informed choices about software development approaches for their projects, with respect to cost, duration and quality |
|                          | S4  | Collect evidence on benefits and limitations of agile software development, understand the strength of evidence of the findings, practical and research implications                                                                                         |
|                          | S5  | Assess the relationship between Agile Software Development (ASD) and Open Source Software Development (OSSD).                                                                                                                                                |
|                          | S6  | Characterize reconciliation among the plan-driven, agile, and free/open source software models of software development.                                                                                                                                      |
|                          | S8  | Identify the origin, prevalence, benefits, enablers, and problems of rapid releases.                                                                                                                                                                         |
|                          | S12 | Evaluate, synthesize, and present the existing findings that will give the latest state of research on applying agile software development method to embedded software development.                                                                          |
|                          | S13 | Present an detailed view about uses of reference architectures in agile methodologies                                                                                                                                                                        |
|                          | S14 | Identify the software practices usually used into the context of agile approaches for the development of software.                                                                                                                                           |
|                          | S15 | Provide insight into the Kanban approach and its elements (concepts, principles, practices, techniques, and tools) that have been empirically reported by scholars and practitioners.                                                                        |
|                          | S16 | Identify the characteristics of agile project management in organizations using agile methods and maturity models; regarding support approaches employed; from the viewpoint of researchers; in academic and industrial context.                             |
|                          | S17 | Provide empirical support for a proposal of a methodology for integration of User Centered Design (UCD) and Agile, identifying most common practices and artefacts used.                                                                                     |
|                          | S18 | Identify what barriers have been dealt with, and what challenges have to be addressed in the near future to apply Agile Product Line Engineering (APLE) to the software industry.                                                                            |
|                          | S20 | Find out to what experiences there are of Model-driven Agile Development (MAD), from an empirical context.                                                                                                                                                   |
|                          | S22 | Identify, synthesize, and present the findings reported about using Scrum practices in Global Software Development (GSD) to date.                                                                                                                            |
|                          | S23 | Systematically reviewing and summarizing the existing research literature and investigating which Agile practices have been used effectively in Global Software Engineering (GSE) contexts.                                                                  |
|                          | S28 | Present a detailed view of how agile methods have been used in the development of embedded systems, and to describe their benefits, challenges, and limitations.                                                                                             |
|                          | S29 | Identify how Agile create business values                                                                                                                                                                                                                    |
|                          | S30 | Identify various challenging factors that restrict Agile and User Centred Design Integration (AUCDI) and explore the proposed practices to deal with them.                                                                                                   |
|                          | S31 | Gain a comprehensive understanding of the various factors that impact the sustained usage of agile methods.                                                                                                                                                  |
|                          | S33 | Evaluate, synthesize, and present results on the use of the Capability Maturity Model Integration (CMMI) in combination with agile software development.                                                                                                     |
|                          | S35 | Examine industrial surveys published in 2011 and 2012, determine the extent to which we could trust their reported high rates of agile method usage and provide recommendations on how quality of research could be improved in the future.                  |
|                          | S37 | Provide an overview of studies within knowledge management in agile projects, what kind of concepts have been explored, what the main findings were and what were the research method                                                                        |
|                          | S38 | Amass current knowledge about agile adoption and to identify essential future research issues for empirical studies, especially on agile in the large settings.                                                                                              |
| Survey the literature    | S39 | Identify technical and organizational challenges of Service Oriented Architecture (SOA) and cloud in Agile context                                                                                                                                           |
|                          | S1  | Investigate the status of Kanban in software development, in terms of its presence in existing literature.                                                                                                                                                   |
|                          | S25 | Review the literature of actual use of software metrics in the context of agile software development.                                                                                                                                                        |
|                          | S27 | Review the current research literature on effort estimation in agile, iterative and incremental software projects (AIISPs) and evidences about common trends and gaps                                                                                        |
|                          | S31 | Provide a synthesis of relevant studies in this issue in order to identify the scope of current research and the shortcomings that need to be addressed in the future.                                                                                       |
|                          | S34 | Investigate the extent to which agile practices have been used in scientific software projects. Second, we aim to investigate the impact on testing and requirements activities in projects with agile practices.                                            |
|                          | S36 | Provide a detailed overview of the state of the art in the area of effort estimation in ASD.                                                                                                                                                                 |
|                          | S40 | Evaluates the potential of agile methods and techniques to address the challenges of Model-Driven Modernization.                                                                                                                                             |
|                          | S41 | Identify testing related problems in the context of the automotive software domain and solutions that have been proposed and applied in an industrial context.                                                                                               |

**Appendix A. Aims of secondary studies****References**

Table A.27

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